

CANONICAL–MICROCANONICAL POSITIVITY AND PHASE GEOMETRY FOR A BESSEL $_{-0}F_1$ MATRIX TURÁN DETERMINANT

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ABSTRACT. Order log-concavity and the argument Turán inequality are two classical positivity directions for modified Bessel functions. We prove that they admit a sharp matrix coupling: the symmetric matrix formed from the order curvature, the mixed order–argument derivative, and the classical Turánian is positive definite throughout the full range $\nu > -1$, $z > 0$, after a trigamma correction forced by the vacuum limit. Passing to a reciprocal-gamma series identifies the coefficient matrices with Schur deficits of conditioned discrete Bessel ensembles; mixed-determinant polarization then localizes the endpoint defect to one fiber and one determinant degree. This representation separates coefficientwise and pointwise positivity into microcanonical and canonical phase boundaries controlled by the vacuum, one-particle, and thermodynamic sectors. Equivalence of ensembles then matches the high-degree coefficient walls with the large-argument Bessel boundary, with cubic agreement at a unique critical point.

1. INTRODUCTION

Turán inequalities for modified Bessel functions appear as inequalities between neighboring orders, monotonicity of consecutive-order ratios, and bounds for logarithmic derivatives [4, 6, 43]. A complementary literature develops log-concavity and coefficientwise Turán inequalities for hypergeometric and reciprocal-gamma series [21, 22, 23, 25, 26]. Two scalar directions are therefore already understood. The order curvature is positive because I_ν is strictly log-concave in ν , while the argument direction below is a classical shifted modified-Bessel Turánian. The question here is whether these directions admit a genuinely mixed matrix coupling.

1.1. Main result. Put $\Theta_z = z\partial_z$ and

$$G_\nu = -\partial_\nu^2 \log I_\nu, \quad P_\nu = \partial_\nu(\Theta_z \log I_\nu), \quad H_\nu = 2(\Theta_z \log I_\nu - \nu) - \Theta_z^2 \log I_\nu.$$

The new object is the *Bessel–Schur matrix*

$$(1.1) \quad \mathcal{S}_\nu(z) = \begin{pmatrix} G_\nu(z) & 1 + P_\nu(z) \\ 1 + P_\nu(z) & H_\nu(z) + 4/\psi_1(\nu + 1) \end{pmatrix}.$$

We prove that $\mathcal{S}_\nu(z) \succ 0$ for every $\nu > -1$ and $z > 0$. The diagonal inequalities themselves are classical; the new statement is the strict Schur coupling through the mixed order–argument derivative. The trigamma correction is forced by the vacuum boundary: as $z \downarrow 0$, the matrix tends to a rank-one matrix, and changing the correction by a scalar factor moves its limiting determinant through zero. Thus the endpoint is not an arbitrary strengthening of two scalar inequalities but the unique closure of their mixed Schur deficit. Theorem 2.5 and Proposition 2.7 state the inequality and its minimality; Section 3 records the classical scalar identifications and makes this separation precise.

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1.2. Fixed-total fibers and the three-resolution hierarchy. The proof is most naturally carried out after passing to the reciprocal-gamma series

$$Z(a, \lambda) = \sum_{k \geq 0} \frac{\lambda^k}{k! \Gamma(a+k)} = \frac{{}_0F_1(; a; \lambda)}{\Gamma(a)}, \quad a = \nu + 1, \quad z = 2\sqrt{\lambda}.$$

Parameter and Euler derivatives of Z form a two-parameter matrix $\mathcal{T}_{\kappa, \tau}$ whose determinant is carried to the Bessel determinant by a positive diagonal congruence. A reciprocal-gamma convolution decomposes this matrix into fixed-degree fibers $M_m^{(\kappa, \tau)}$. At the endpoint, M_0 has rank one, every M_m with $m \geq 2$ is positive definite, and M_1 is the only fiber that can be indefinite. Mixed-determinant polarization then localizes its self-interaction to determinant degree two. The infinite coefficientwise problem therefore reduces to one finite defect.

The same decomposition has a probabilistic meaning that organizes the paper. The fibers are Schur-deficit matrices for two independent discrete Bessel counts conditioned on their total; determinant coefficients are obtained by conditioning four copies on their grand total; and the pointwise Bessel determinant is the canonical average of these fixed-total sectors. A single Schur deficit is therefore being viewed at three resolutions:

$$\text{two-copy fiber} \longrightarrow \text{four-copy determinant sector} \longrightarrow \text{canonical Bessel determinant}.$$

In particular, coefficientwise and pointwise positivity are respectively the microcanonical and canonical positivity problems for the same ensemble.

1.3. Phase geometry and thermodynamic matching. The two deformation parameters expose the distinction between these positivity notions. The vacuum sector gives the pointwise boundary $\tau = 1$, the one-particle sector gives the exposed positive-degree coefficient boundary $\tau = \tau_{\text{cw}}(a, \kappa)$, and the thermodynamic sector gives the deformation boundary $\kappa = 1$. These are the zero-, one-, and infinite-particle contacts of one sector family. Between the coefficientwise and pointwise boundaries, every positive microcanonical sector is nonnegative while the vacuum sector is negative, and the canonical determinant has exactly one positive zero.

The individual coefficient walls form an affine fan near the thermodynamic contact. On the canonical side, the four-copy total concentrates at $N \sim 2z$. The usual coefficient saddle equation is exactly the equation that tunes the canonical ensemble to a prescribed total, so the agreement between high-degree coefficient asymptotics and the large-argument Bessel asymptotics is equivalence of ensembles rather than a separate saddle-point coincidence. The matching persists through successive critical cancellations and becomes cubic at a unique point of the parameter family.

1.4. Organization. Section 2 develops the reciprocal-gamma framework, derives its Bessel correspondence, and states the coefficientwise, pointwise, and critical-scaling theorems, together with the global phase diagram. Section 3 treats the classical scalar Bessel directions. Section 4 derives the reciprocal-gamma convolution, the fixed-total coefficient matrices it produces, and their canonical–microcanonical interpretation. Section 5 proves endpoint positivity by Gram structure and finite-defect localization. Section 6 proves the exposed coefficient wall and the resulting phase geometry, and Section 7 develops the wall fan, equivalence of ensembles, and higher-order critical matching. Section 8 places the mechanism relative to neighboring positivity theories, and Section 9 collects the structural consequences and open problems. Two appendices contain the lengthy exposed-wall verification and the continuation beyond the positive-series domain.

2. RECIPROCAL-GAMMA FORMULATION AND POSITIVITY PHASE DIAGRAM

Throughout, $a > 0$, $\lambda \geq 0$, and $\mathbb{N}_0 = \{0, 1, 2, \dots\}$. Put

$$(2.1) \quad Z(a, \lambda) = \sum_{k=0}^{\infty} \frac{\lambda^k}{k! \Gamma(a+k)} = \frac{{}_0F_1(; a; \lambda)}{\Gamma(a)}.$$

We first record the analyticity needed here and in Appendix B.

Lemma 2.1. *The series (2.1) and all of its fixed-order parameter and Euler derivatives converge locally uniformly on $\mathbb{C}^2$. Hence Z and those derivatives are entire in (a, λ) .*

Proof. Each summand of (2.1) is entire in a . Uniform Stirling estimates for a in compact sets, together with Cauchy estimates for derivatives of $1/\Gamma$, give locally uniform convergence of the series and of every fixed-order parameter and Euler derivative on compact subsets of $\mathbb{C}^2$. $\square$

Write $\Theta_\lambda = \lambda \partial_\lambda$, $g = \psi_1(a)$, and $\Theta_z = z \partial_z$.

For $\kappa, \tau \in \mathbb{R}$, define

$$(2.2) \quad A = Z_a^2 - Z Z_{aa},$$

$$(2.3) \quad B = Z^2 + Z Z_{a\Theta_\lambda} - Z_a Z_{\Theta_\lambda},$$

$$(2.4) \quad C_{\kappa, \tau} = \tau Z^2 + g(\kappa Z Z_{\Theta_\lambda} - Z Z_{\Theta_\lambda \Theta_\lambda} + Z_{\Theta_\lambda}^2),$$

$$(2.5) \quad \mathcal{T}_{\kappa, \tau} = \begin{pmatrix} A & \sqrt{g} B \\ \sqrt{g} B & C_{\kappa, \tau} \end{pmatrix}, \quad \Delta^{(\kappa, \tau)} = \det \mathcal{T}_{\kappa, \tau}.$$

By a *matrix Turán determinant* we mean this confluent determinant assembled from parameter and Euler derivatives, rather than a determinant of parameter-shifted evaluations; the distinction from classical Turán and total-positivity constructions is discussed in Section 8. On $\tau = 1$, write $C_\kappa, \mathcal{T}_\kappa, \Delta^{(\kappa)}$; at $(\kappa, \tau) = (1, 1)$, write $C, \mathcal{T}, \Delta$.

Remark 2.2. Writing $L = \log Z$,

$$\frac{A}{Z^2} = -L_{aa}, \quad \frac{B}{Z^2} = 1 + L_{a\Theta_\lambda}, \quad \frac{C_{\kappa, \tau}}{g Z^2} = \frac{\tau}{g} + \kappa L_{\Theta_\lambda} - L_{\Theta_\lambda \Theta_\lambda}.$$

Thus κ deforms the Euler direction and τ varies an independent Schur correction.

Set

$$(2.6) \quad P = A Z^2, \quad Q = g A Z Z_{\Theta_\lambda}.$$

At determinant level the deformations are affine:

$$(2.7) \quad \Delta^{(\kappa, \tau)} = \Delta + (\kappa - 1)Q + (\tau - 1)P.$$

Both P and Q have strictly positive coefficients in every positive degree.

A function on $(0, \infty)$ is *strictly absolutely monotone* if it and all its derivatives are strictly positive there.

Theorem 2.3 (Sharp endpoint coefficientwise classification). *For each $a > 0$,*

$$\Delta^{(\kappa)}(a, \lambda) = \sum_{n \geq 1} \Delta_n^{(\kappa)}(a) \lambda^n, \quad \Delta_n^{(\kappa)}(a) > 0 \quad (n \geq 1),$$

holds if and only if $\kappa \geq 1$. Consequently $\Delta^{(\kappa)}(a, \cdot)$ is strictly absolutely monotone and $\mathcal{T}_\kappa(a, \lambda) \succ 0$ for $\kappa \geq 1, \lambda > 0$.

The constant coefficient satisfies

$$(2.8) \quad [\lambda^0] \Delta^{(\kappa, \tau)} = \frac{g(\tau - 1)}{\Gamma(a)^4}.$$

Define

$$(2.9) \quad \tau_{\text{cw}}(a, \kappa) = \frac{ag(2a - 1) - \frac{1}{2}(\kappa - 1)a^2 g^2}{2a^2 g - 1}.$$

The denominator is positive because $a^2 g = 1 + a^2 \sum_{r \geq 1} (a + r)^{-2} > 1$.

Theorem 2.4 (Two-parameter coefficientwise phase diagram). *Fix $a > 0$. All positive-degree coefficients of $\Delta^{(\kappa, \tau)}(a, \lambda)$ are nonnegative if and only if*

$$\kappa \geq 1, \quad \tau \geq \tau_{\text{cw}}(a, \kappa).$$

They are all strictly positive if and only if

$$\kappa \geq 1, \quad \tau > \tau_{\text{cw}}(a, \kappa).$$

On the affine boundary $\tau = \tau_{\text{cw}}(a, \kappa)$, the degree-one coefficient vanishes and every coefficient of degree $n \geq 2$ is strictly positive. Moreover $\tau_{\text{cw}}(a, \kappa) < 1$ for every $\kappa \geq 1$.

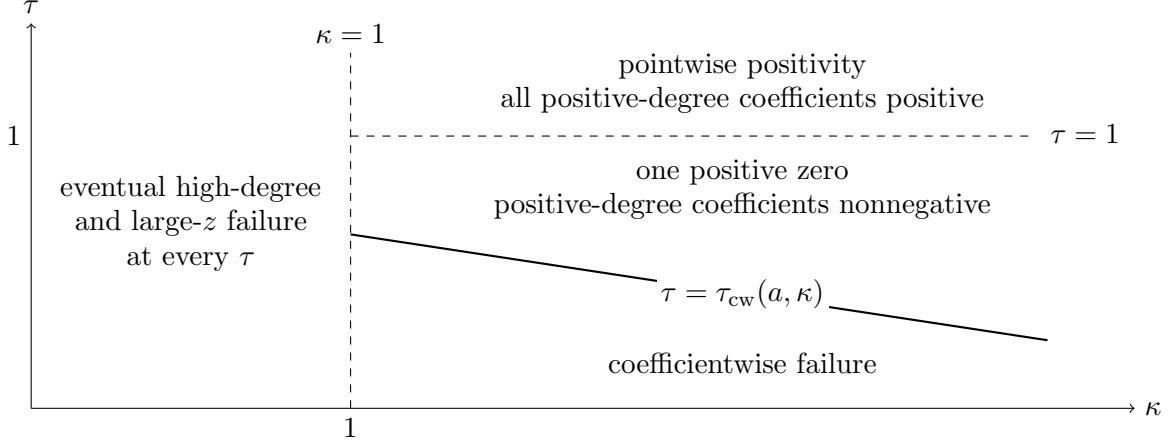


FIGURE 1. Schematic global phase diagram for fixed $a > 0$. The horizontal boundary $\tau = 1$ is the vacuum contact and pointwise boundary, the vertical boundary $\kappa = 1$ is the thermodynamic contact, and the affine line $\tau = \tau_{\text{cw}}(a, \kappa)$ is the exposed one-particle wall and lower boundary for nonnegative positive-degree coefficients. For $\kappa < 1$, both notions fail at every τ . Only the global phase boundaries are shown; the interior coefficient-wall fan is displayed separately in Figure 3.

The Bessel form follows from the standard identity

$$(2.10) \quad I_{a-1}(2\sqrt{\lambda}) = \lambda^{(a-1)/2} Z(a, \lambda)$$

[32, eq. 10.25.2], [43, §3.7]. Put $a = \nu + 1$, $z = 2\sqrt{\lambda}$, and define

$$(2.11) \quad G_\nu = -\partial_\nu^2 \log I_\nu,$$

$$(2.12) \quad P_\nu = \partial_\nu(\Theta_z \log I_\nu),$$

$$(2.13) \quad H_\nu^{(\kappa)} = 2\kappa(\Theta_z \log I_\nu - \nu) - \Theta_z^2 \log I_\nu.$$

Throughout, a subscript ν marks a Bessel-side quantity: P_ν is not P , and the ratio R_ν of Section 3 is not the pair total R of Section 4. At $\kappa = 1$, write $H_\nu = H_\nu^{(1)}$. Define

$$(2.14) \quad D_\nu^{(\kappa, \tau)} = G_\nu \left(H_\nu^{(\kappa)} + \frac{4\tau}{\psi_1(\nu+1)} \right) - (1 + P_\nu)^2.$$

At the endpoint write

$$(2.15) \quad D_\nu = D_\nu^{(1,1)} = G_\nu \left(H_\nu + \frac{4}{\psi_1(\nu+1)} \right) - (1 + P_\nu)^2.$$

The identity (2.10) gives the differential correspondence needed to transfer the reciprocal-gamma determinant to the Bessel side. With $L = \log Z$, $z = 2\sqrt{\lambda}$, $\nu = a - 1$, and $\Theta_z = 2\Theta_\lambda$,

$$(2.16) \quad \Theta_z \log I_\nu = \nu + 2L_{\Theta_\lambda}, \quad P_\nu = 1 + 2L_{a\Theta_\lambda},$$

$$(2.17) \quad G_\nu = -L_{aa}, \quad H_\nu^{(\kappa)} = 4\kappa L_{\Theta_\lambda} - 4L_{\Theta_\lambda \Theta_\lambda}.$$

Together with Remark 2.2, this gives

$$(2.18) \quad \begin{pmatrix} G_\nu & 1 + P_\nu \\ 1 + P_\nu & H_\nu^{(\kappa)} + 4\tau/g \end{pmatrix} = \frac{1}{Z^2} \begin{pmatrix} 1 & 0 \\ 0 & 2/\sqrt{g} \end{pmatrix} \mathcal{T}_{\kappa, \tau}(a, \lambda) \begin{pmatrix} 1 & 0 \\ 0 & 2/\sqrt{g} \end{pmatrix}.$$

Taking determinants yields

$$(2.19) \quad D_{a-1}^{(\kappa, \tau)}(2\sqrt{\lambda}) = \frac{4\Delta^{(\kappa, \tau)}(a, \lambda)}{gZ(a, \lambda)^4}.$$

At $(\kappa, \tau) = (1, 1)$, one also has

$$(2.20) \quad \frac{A}{Z^2} = -L_{aa}, \quad \frac{B}{Z^2} = 1 + L_{a\Theta_\lambda}, \quad \frac{C}{gZ^2} = \frac{1}{g} + L_{\Theta_\lambda} - L_{\Theta_\lambda\Theta_\lambda},$$

and

$$(2.21) \quad D_{a-1}(2\sqrt{\lambda}) = \frac{4\Delta(a, \lambda)}{gZ(a, \lambda)^4}.$$

Theorem 2.5 (Mixed Bessel–Schur inequality). *For $\nu > -1$ and $z > 0$,*

$$(2.22) \quad G_\nu(z) \left(H_\nu(z) + \frac{4}{\psi_1(\nu+1)} \right) > (1 + P_\nu(z))^2.$$

Corollary 2.6. *For $\nu > -1$, $z > 0$,*

$$\begin{pmatrix} G_\nu(z) & 1 + P_\nu(z) \\ 1 + P_\nu(z) & H_\nu(z) + 4/\psi_1(\nu+1) \end{pmatrix} \succ 0.$$

As $z \downarrow 0$, it extends continuously to the rank-one matrix $\begin{pmatrix} g & 2 \\ 2 & 4/g \end{pmatrix}$.

Proposition 2.7. *Fix $\nu > -1$, put $a = \nu + 1$, and let $\kappa, \tau \in \mathbb{R}$. Then $D_\nu^{(\kappa, \tau)}(z) > 0$ for every $z > 0$ if and only if*

$$\kappa \geq 1, \quad \tau \geq 1.$$

Equivalently, the associated Bessel–Schur matrix is positive definite for all $z > 0$ exactly in this quadrant. If $\kappa \geq 1$ and $\tau_{\text{cw}}(a, \kappa) \leq \tau < 1$, then $D_\nu^{(\kappa, \tau)}$ has exactly one zero on $(0, \infty)$.

For the critical scaling, put

$$(2.23) \quad q_n(a) = [\lambda^n]Q(a, \lambda) > 0, \quad \kappa_n(a) = 1 - \frac{\Delta_n(a)}{q_n(a)},$$

and

$$(2.24) \quad c(a) = \frac{4}{\psi_1(a)} - 4a + \frac{7}{2} = \frac{4}{\psi_1(\nu+1)} - 4\nu - \frac{1}{2}.$$

Theorem 2.8. *For every fixed $a > 0$,*

$$(2.25) \quad 1 - \kappa_n(a) = \frac{c(a)}{n} + \frac{\frac{1}{12} - \frac{2}{\psi_1(a)}}{n^2} + O_a(n^{-3}).$$

The estimate is uniform for a in every compact subinterval of $(0, \infty)$. Moreover

$$(2.26) \quad c(a) = \lim_{z \rightarrow \infty} zD_{a-1}(z) = \lim_{n \rightarrow \infty} n(1 - \kappa_n(a)),$$

$$(2.27) \quad \frac{3}{2} < c(a) < \frac{7}{2},$$

with $c(a)$ strictly decreasing from $7/2$ to $3/2$ as a runs from 0 to ∞ , and $\kappa_n(a)$ is eventually strictly increasing. For $\kappa < 1$, define

$$N_a(\kappa) = \min\{n \geq 1 : \Delta_n^{(\kappa)}(a) < 0\}.$$

Then

$$(2.28) \quad N_a(\kappa) \sim \frac{c(a)}{1 - \kappa} \quad (\kappa \uparrow 1).$$

For the full deformation wall, put

$$(2.29) \quad p_n(a) = [\lambda^n]P(a, \lambda), \quad \kappa_{n,\tau}(a) = 1 - \frac{\Delta_n(a) + (\tau - 1)p_n(a)}{q_n(a)}.$$

For $n \geq 1$, the same affine coefficient defines the individual degree wall

$$(2.30) \quad \tau_n(a, \kappa) = 1 - \frac{\Delta_n(a) + (\kappa - 1)q_n(a)}{p_n(a)}.$$

Define

$$(2.31) \quad c_\tau(a) = \frac{4\tau}{g} - 4a + \frac{7}{2}, \quad d_\tau(a) = \frac{1}{12} - \frac{2\tau}{g}, \quad g = \psi_1(a).$$

Theorem 2.9. *For fixed $a > 0$ and $\tau \in \mathbb{R}$,*

$$(2.32) \quad 1 - \kappa_{n,\tau}(a) = \frac{c_\tau(a)}{n} + \frac{d_\tau(a)}{n^2} + O_{a,\tau}(n^{-3}).$$

The estimate is locally uniform in $(a, \tau) \in (0, \infty) \times \mathbb{R}$. On the Bessel side,

$$(2.33) \quad D_{a-1}^{(1,\tau)}(z) = \frac{c_\tau(a)}{z} + \frac{e_\tau(a)}{z^2} + O_{a,\tau}(z^{-3}), \quad e_\tau(a) = \frac{2\tau}{g} - 3(a - 1) - \frac{1}{3}.$$

The high-degree walls accumulate at

$$(2.34) \quad \tau_\infty(a) = g \left(a - \frac{7}{8} \right),$$

for which $c_{\tau_\infty} = 0$ and

$$(2.35) \quad 1 - \kappa_{n,\tau_\infty}(a) = \frac{2(11/12 - a)}{n^2} + O_a(n^{-3}), \quad D_{a-1}^{(1,\tau_\infty)}(z) = \frac{11/12 - a}{z^2} + O_a(z^{-3}).$$

Moreover

$$(2.36) \quad \tau_\infty(a) < \tau_{\text{cw}}(a, 1) < 1.$$

Theorem 2.10 (Cubic wall-fan multicritical point). *Let*

$$(2.37) \quad a_{\text{mc}} = \frac{11}{12}, \quad \nu_{\text{mc}} = -\frac{1}{12}, \quad g_{\text{mc}} = \psi_1(11/12), \quad \tau_{\text{mc}} = \frac{g_{\text{mc}}}{24}.$$

Then the first two wall jets vanish and

$$(2.38) \quad 1 - \kappa_{n,\tau_{\text{mc}}}(a_{\text{mc}}) = \frac{3}{8n^3} + O(n^{-4}),$$

while

$$(2.39) \quad D_{\nu_{\text{mc}}}^{(1,\tau_{\text{mc}})}(z) = \frac{3}{32z^3} + O(z^{-4}).$$

Thus the coefficient–argument matching persists through the third critical jet under $n \sim 2z$.

Corollary 2.11. *At $\kappa = 1$,*

$$(2.40) \quad \tau_n(a, 1) = \tau_\infty(a) + \frac{g}{2n} \left(a - \frac{11}{12} \right) + O_a(n^{-2}).$$

Thus the high-degree walls approach $\tau_\infty(a)$ eventually from below when $a < 11/12$ and from above when $a > 11/12$. At the switching value,

$$(2.41) \quad \tau_n(11/12, 1) = \tau_{\text{mc}} - \frac{3g_{\text{mc}}}{32n^2} + O(n^{-3}).$$

For fixed κ ,

$$(2.42) \quad \tau_n(a, \kappa) = \tau_\infty(a) - \frac{g}{4}(\kappa - 1)n + O_{a,\kappa}(1),$$

so the wall heights tend to $-\infty$, $\tau_\infty(a)$, or $+\infty$ according as $\kappa > 1$, $\kappa = 1$, or $\kappa < 1$. More generally, for every fixed $x \in \mathbb{R}$,

$$(2.43) \quad \tau_n \left(a, 1 - \frac{x}{n} \right) \longrightarrow \tau_\infty(a) + \frac{gx}{4}.$$

The resulting global regions are summarized schematically in Figure 1; the individual coefficient walls are resolved in Figure 3.

3. CLASSICAL SCALAR BESSEL DIRECTIONS

The diagonal entries of the Bessel–Schur matrix are classical scalar quantities, and no positivity result of the present paper is used in identifying them.

Put

$$(3.1) \quad R_\nu(z) = \frac{I_{\nu+1}(z)}{I_\nu(z)}, \quad r_\nu(z) = zR_\nu(z) = \Theta_z \log I_\nu(z) - \nu.$$

Both letters denote first-kind quantities here.¹

Proposition 3.1. *For $\nu > -1$, $z > 0$, and $\kappa \in \mathbb{R}$,*

$$(3.2) \quad H_\nu^{(\kappa)} = r_\nu^2 + 2(\nu + \kappa)r_\nu - z^2 = 2\kappa r_\nu - zr'_\nu.$$

Consequently

$$(3.3) \quad H_\nu^{(\kappa)}(z) > 0 \iff R_\nu(z) > \frac{z}{\nu + \kappa + \sqrt{(\nu + \kappa)^2 + z^2}}.$$

At $\kappa = 1$,

$$(3.4) \quad H_\nu(z) = z^2 R_\nu(z)(R_\nu(z) - R_{\nu+1}(z)) = z^2 \frac{I_{\nu+1}(z)^2 - I_\nu(z)I_{\nu+2}(z)}{I_\nu(z)^2}.$$

In particular, $H_\nu > 0$ is equivalent both to the shifted modified-Bessel Turán inequality and to the classical Amos-type lower bound

$$(3.5) \quad R_\nu(z) > \frac{z}{\nu + 1 + \sqrt{(\nu + 1)^2 + z^2}}.$$

Finally,

$$(3.6) \quad H_\nu^{(\kappa)}(z) > 0 \text{ for every } z > 0 \iff \kappa \geq 1.$$

Proof. The modified Bessel equation [32, eq. 10.25.1] gives

$$\Theta_z^2 \log I_\nu = z^2 + \nu^2 - (\Theta_z \log I_\nu)^2 = z^2 - 2\nu r_\nu - r_\nu^2.$$

Substitution in (2.13) gives the first expression in (3.2); since $\Theta_z^2 \log I_\nu = \Theta_z r_\nu = zr'_\nu$, the second follows as well. Since $I_\nu(z) > 0$ for $\nu > -1$ and $z > 0$, one has $r_\nu > 0$. The quadratic $r^2 + 2(\nu + \kappa)r - z^2$ has one positive and one negative root, so $H_\nu^{(\kappa)} > 0$ exactly when r_ν exceeds the positive root

$$\sqrt{(\nu + \kappa)^2 + z^2} - (\nu + \kappa) = \frac{z^2}{\nu + \kappa + \sqrt{(\nu + \kappa)^2 + z^2}},$$

which proves (3.3). The recurrence $R_{\nu+1} = R_\nu^{-1} - 2(\nu + 1)/z$ [43, §3.71] gives (3.4). Positivity at $\kappa = 1$ is the classical modified-Bessel Turán inequality on $\nu > -1$ [4, Thm. 2.1], [6, §1]; equivalently it is the Amos-type quotient inequality (3.5) [2, 19, 30, 39]. For $\kappa \geq 1$,

$$H_\nu^{(\kappa)} = H_\nu + 2(\kappa - 1)r_\nu > 0.$$

Conversely the ascending series gives $r_\nu(z) = z^2/(2(\nu + 1)) + O(z^4)$, hence

$$H_\nu^{(\kappa)}(z) = \frac{\kappa - 1}{\nu + 1} z^2 + O(z^4),$$

which is negative near zero when $\kappa < 1$. This proves (3.6). $\square$

¹The capital R_ν agrees with [19]; the lowercase agrees with neither of the two works that use it. Both [39] and [19] write r_ν for $z^{-1}I_\nu/I_{\nu-1}$, which is $R_{\nu-1}/z$ and not the zR_ν used here. In [39], R_ν is in addition the second-kind ratio $z^{-1}K_\nu/K_{\nu+1}$.

Remark 3.2. The correction $4/g$ is not required for scalar positivity of the second diagonal direction: $H_\nu > 0$ already holds by Proposition 3.1. Its role is matrix-theoretic. As $z \downarrow 0$,

$$\begin{pmatrix} G_\nu & 1 + P_\nu \\ 1 + P_\nu & H_\nu + 4\tau/g \end{pmatrix} \longrightarrow \begin{pmatrix} g & 2 \\ 2 & 4\tau/g \end{pmatrix},$$

whose determinant is $4(\tau - 1)$. Thus $4/g = 2^2/g$ is exactly the Schur amount needed to close the mixed deficit at the rank-one boundary.

4. RECIPROCAL-GAMMA CONVOLUTION AND CANONICAL-MICROCANONICAL STRUCTURE

The coefficient calculation rests on a two-parameter Chu–Vandermonde identity.

Lemma 4.1 (Asymmetric reciprocal-gamma convolution). *Let $m \in \mathbb{N}_0$ and let $\alpha, \beta > 0$. Then*

$$(4.1) \quad \sum_{i=0}^m \frac{1}{i!(m-i)!\Gamma(\alpha+i)\Gamma(\beta+m-i)} = \frac{(\alpha+\beta+m-1)_m}{m!\Gamma(\alpha+m)\Gamma(\beta+m)}.$$

Both sides are entire in (α, β) .

Proof. Multiply the left-hand side by $m!\Gamma(\alpha+m)\Gamma(\beta+m)$. Using

$$\frac{\Gamma(\alpha+m)}{\Gamma(\alpha+i)} = (\alpha+i)_{m-i} = \frac{(\alpha)_m}{(\alpha)_i}$$

and

$$\frac{\Gamma(\beta+m)}{\Gamma(\beta+m-i)} = (\beta+m-i)_i = (-1)^i(1-\beta-m)_i,$$

the resulting sum is

$$(\alpha)_m {}_2F_1(-m, 1-\beta-m; \alpha; 1).$$

Chu–Vandermonde gives $(\alpha+\beta+m-1)_m$. Both sides of (4.1) are entire in (α, β) , being finite sums and products of the entire function $1/\Gamma$ and a polynomial, so agreement on $\{\alpha, \beta > 0\}$ extends the identity to $\mathbb{C}^2$ by the identity theorem in each variable separately. $\square$

Set

$$S_m = S_m(a) = \frac{(2a+m-1)_m}{m!\Gamma(a+m)^2} > 0.$$

Taking $\alpha = \beta = a$ in Lemma 4.1 gives $[\lambda^m]Z(a, \lambda)^2 = S_m$, where $[\lambda^m]$ extracts the coefficient of λ^m . Unsubscripted α, β always denote the parameters of Lemma 4.1; the subscripted families of Theorem 4.2 below are coefficient sequences.

Theorem 4.2. *There are expansions*

$$(4.2) \quad A = \sum_{m \geq 0} S_m \alpha_m \lambda^m, \quad B = \sum_{m \geq 0} S_m \beta_m \lambda^m, \quad C_{\kappa, \tau} = \sum_{m \geq 0} S_m \gamma_m^{(\kappa, \tau)} \lambda^m,$$

where

$$(4.3) \quad \alpha_m = \psi_1(a+m),$$

$$(4.4) \quad \beta_0 = 1, \quad \beta_m = \frac{2a+m-2}{2(a+m-1)} \quad (m \geq 1),$$

and

$$(4.5) \quad \gamma_m^{(\kappa, \tau)} = \tau + g c_m^{(\kappa)},$$

with

$$(4.6) \quad c_0^{(\kappa)} = 0, \quad c_1^{(\kappa)} = \frac{\kappa-1}{2},$$

and, for $m \geq 2$,

$$(4.7) \quad c_m^{(\kappa)} = \frac{\kappa m}{2} - \frac{m(2a+m-2)}{2(2a+2m-3)} = \frac{m[(\kappa-1)(2a+2m-3) + m-1]}{2(2a+2m-3)}.$$

At the endpoint $\kappa = 1$,

$$c_0 = c_1 = 0, \quad c_m = \frac{m(m-1)}{2(2a+2m-3)} \quad (m \geq 2).$$

Consequently,

$$(4.8) \quad \mathcal{T}_{\kappa,\tau}(a, \lambda) = \sum_{m \geq 0} S_m M_m^{(\kappa,\tau)} \lambda^m, \quad M_m^{(\kappa,\tau)} = \begin{pmatrix} \alpha_m & \sqrt{g} \beta_m \\ \sqrt{g} \beta_m & \tau + g c_m^{(\kappa)} \end{pmatrix}.$$

At the endpoint $(\kappa, \tau) = (1, 1)$, this becomes

$$(4.9) \quad \mathcal{T}(a, \lambda) = \sum_{m \geq 0} S_m M_m \lambda^m, \quad M_m = \begin{pmatrix} \alpha_m & \sqrt{g} \beta_m \\ \sqrt{g} \beta_m & 1 + g c_m \end{pmatrix}.$$

Proof. For the first coefficient, define

$$F_m(\delta) = \sum_{i=0}^m \frac{1}{i!(m-i)!\Gamma(a+\delta+i)\Gamma(a-\delta+m-i)}.$$

By Lemma 4.1,

$$(4.10) \quad F_m(\delta) = \frac{(2a+m-1)_m}{m!\Gamma(a+m+\delta)\Gamma(a+m-\delta)}.$$

On the one hand,

$$(4.11) \quad \partial_\delta^2 [Z(a+\delta, \lambda)Z(a-\delta, \lambda)]|_{\delta=0} = 2ZZ_{aa} - 2Z_a^2 = -2A.$$

On the other hand, $F'_m(0) = 0$ and the logarithmic second derivative of (4.10) is $-2\psi_1(a+m)$ at $\delta = 0$, so

$$(4.12) \quad F''_m(0) = -2S_m\psi_1(a+m).$$

Comparing the coefficient of λ^m in (4.11) with (4.12) gives $[\lambda^m]A = S_m\psi_1(a+m)$, proving (4.3).

For $m = 0$, one has $[\lambda^0]B = Z(a, 0)^2 = S_0$, so $\beta_0 = 1$. Assume henceforth that $m \geq 1$. Let $\mathbf{l}$ be the index of the first factor in $Z(a+\delta, \lambda)Z(a-\delta, \lambda) = \sum_m F_m(\delta)\lambda^m$, so that $[\lambda^m]$ weights i by

$$w_i(\delta) = [i!(m-i)!\Gamma(a+\delta+i)\Gamma(a-\delta+m-i)]^{-1}.$$

For real δ with $|\delta| < a$, write $\mathbb{E}_\delta$ for expectation under the probability weights $w_i(\delta)/F_m(\delta)$, and put $\mathbb{E} = \mathbb{E}_0$, $\mathbf{D} = m - 2\mathbf{l}$, and $\mathbf{J} = m - \mathbf{l}$. Applying Lemma 4.1 to $\sum_i i w_i(\delta)$, with $(\alpha, \beta) = (a+1+\delta, a-\delta)$ and $m-1$ in place of m , gives

$$\sum_{i=0}^m i w_i(\delta) = \frac{(2a+m-1)_{m-1}}{(m-1)!\Gamma(a+m+\delta)\Gamma(a+m-1-\delta)}.$$

Dividing by (4.10) yields

$$\mathbb{E}_\delta \mathbf{l} = \frac{m(a+m-1-\delta)}{2(a+m-1)},$$

whence

$$(4.13) \quad \mathbb{E}_\delta \mathbf{D} = \frac{m\delta}{a+m-1}.$$

The score $\partial_\delta \log w_i(\delta)|_{\delta=0}$ is

$$\Xi = \psi(a+m-\mathbf{l}) - \psi(a+\mathbf{l}),$$

so differentiating (4.13) at $\delta = 0$ gives $\mathbb{E}(\mathbf{D}\Xi) = \partial_\delta \mathbb{E}_\delta \mathbf{D}|_{\delta=0}$, that is

$$(4.14) \quad \mathbb{E}(\mathbf{D}\Xi) = \frac{m}{a+m-1}.$$

Direct coefficient extraction gives

$$\begin{aligned} & [\lambda^m](ZZ_{a\Theta_\lambda} - Z_a Z_{\Theta_\lambda}) \\ &= \sum_{i=0}^m \frac{-(m-i)\psi(a+m-i) + (m-i)\psi(a+i)}{i!(m-i)!\Gamma(a+i)\Gamma(a+m-i)} = -S_m \mathbb{E}(\mathbf{J}\Xi). \end{aligned}$$

Together with $[\lambda^m]Z^2 = S_m$, this yields

$$\frac{[\lambda^m]B}{S_m} = 1 - \mathbb{E}(\mathbf{J}\Xi) = 1 - \frac{1}{2}\mathbb{E}(\mathbf{D}\Xi),$$

because $\mathbf{J} = (m + \mathbf{D})/2$ and $\mathbb{E}\Xi = 0$ by symmetry. This proves (4.4).

For $C_{\kappa,\tau}$, the term τZ^2 contributes τ after normalization by S_m , while the Euler-derivative terms contribute

$$\kappa \mathbb{E}\mathbf{J} - \mathbb{E}\mathbf{J}^2 + \mathbb{E}(\mathbf{I}\mathbf{J}).$$

Since $\mathbf{I} = (m - \mathbf{D})/2$, $\mathbf{J} = (m + \mathbf{D})/2$, and $\mathbb{E}\mathbf{D} = 0$ by the symmetry $\mathbf{I} \leftrightarrow m - \mathbf{I}$,

$$(4.15) \quad \kappa \mathbb{E}\mathbf{J} - \mathbb{E}\mathbf{J}^2 + \mathbb{E}(\mathbf{I}\mathbf{J}) = \frac{\kappa m}{2} - \frac{1}{2}\mathbb{E}\mathbf{D}^2.$$

For $m \geq 2$, canceling the factors $i(m-i)$ and shifting $j = i - 1$ gives

$$\begin{aligned} S_m(a) \mathbb{E}[\mathbf{I}(m - \mathbf{I})] &= \sum_{i=1}^{m-1} \frac{1}{(i-1)!(m-i-1)!\Gamma(a+i)\Gamma(a+m-i)} \\ &= S_{m-2}(a+1). \end{aligned}$$

Therefore

$$\mathbb{E}[\mathbf{I}(m - \mathbf{I})] = \frac{S_{m-2}(a+1)}{S_m(a)} = \frac{m(m-1)(a+m-1)}{2(2a+2m-3)},$$

so

$$\mathbb{E}\mathbf{D}^2 = \frac{m(2a+m-2)}{2a+2m-3}.$$

Substitution gives (4.7); the cases $m = 0, 1$ are immediate, and adding the τZ^2 contribution proves (4.5). $\square$

Remark 4.3. Theorem 4.2 does not assert $M_m \succeq 0$ for every m : M_1 can be indefinite for small $a > 0$ (see Section 5.1). The general coefficient matrix is $M_m^{(\kappa,\tau)}$ of (4.8); the endpoint matrix M_m means $M_m^{(1,1)}$.

Remark 4.4. Let $e_2 = (0, 1)^T$. The explicit formula for $c_m^{(\kappa)}$ gives, for every $m \geq 0$,

$$(4.16) \quad M_m^{(\kappa,\tau)} = M_m + \left[(\tau - 1) + \frac{gm}{2}(\kappa - 1) \right] e_2 e_2^T.$$

Thus the vacuum fiber $m = 0$ sees only the correction parameter τ , the first nonconstant fiber sees the first affine combination of τ and κ , and the κ -deformation is amplified linearly with the fiber total. This is the coefficient-level origin of the three phase scales.

4.1. Microcanonical Bessel fibers and canonical averaging. For $a > 0$, $\lambda > 0$, let $Y, Y_1, Y_2, \dots$ be independent random variables with

$$(4.17) \quad \mathbb{P}_\lambda(Y = k) = \frac{\lambda^k}{k! \Gamma(a+k) Z(a, \lambda)}, \quad k \in \mathbb{N}_0.$$

Under $a = \nu + 1$, $z = 2\sqrt{\lambda}$, this is the classical discrete Bessel law

$$\mathbb{P}(Y = k) = \frac{(z/2)^{2k+\nu}}{I_\nu(z) k! \Gamma(k + \nu + 1)}$$

studied by Yuan and Kalbfleisch and by Devroye [44, 12].

Put $R = Y_1 + Y_2$. The law (4.17) is a power-series distribution in the classical sense [35]; for a modern account of the correspondence between nonnegative power series and their associated probability laws and moments, see [10]. Fixed-sum conditional characterizations of discrete

distributions are classical [36]. Here the conditioning is used to construct the Bessel Schur-deficit fibers. For fixed m , let $\mathbf{l}_m$ have the conditional law of Y_1 given $R = m$, and set

$$\begin{aligned} D_m &= m - 2\mathbf{l}_m, & \Xi_m &= \psi(a + m - \mathbf{l}_m) - \psi(a + \mathbf{l}_m), \\ \Sigma_m &= \psi_1(a + \mathbf{l}_m) + \psi_1(a + m - \mathbf{l}_m). \end{aligned}$$

Theorem 4.5. *For every $m \in \mathbb{N}_0$,*

$$(4.18) \quad \mathbb{P}_\lambda(R = m) = \frac{S_m \lambda^m}{Z(a, \lambda)^2}, \quad \mathbb{P}(\mathbf{l}_m = i) = \frac{1}{S_m i!(m-i)!\Gamma(a+i)\Gamma(a+m-i)}.$$

The conditional law is independent of λ . Define, near $(u, v) = (0, 0)$,

$$\mathcal{F}_m(u, v) = \sum_{i=0}^m \frac{\exp\{-v(m-2i)/\sqrt{2}\}}{i!(m-i)!\Gamma(a+u/\sqrt{2}+i)\Gamma(a-u/\sqrt{2}+m-i)}, \quad \ell_m = \log \mathcal{F}_m.$$

Then the normalized coefficient matrix

$$\widehat{M}_m^{(\kappa, \tau)} = \text{diag}(1, g^{-1/2}) M_m^{(\kappa, \tau)} \text{diag}(1, g^{-1/2})$$

has the microcanonical differential form

$$(4.19) \quad \widehat{M}_m^{(\kappa, \tau)} = \begin{pmatrix} -\ell_{m,uu} & 1 + \ell_{m,uv} \\ 1 + \ell_{m,uv} & \tau/g + \kappa m/2 - \ell_{m,vv} \end{pmatrix}_{(u,v)=(0,0)}.$$

Equivalently,

$$(4.20) \quad \alpha_m = \frac{1}{2} \mathbb{E}(\Sigma_m - \Xi_m^2), \quad \beta_m = 1 - \frac{1}{2} \mathbb{E}(D_m \Xi_m), \quad c_m^{(\kappa)} = \frac{\kappa m}{2} - \frac{1}{2} \mathbb{E} D_m^2.$$

Since $\mathbb{E} \Xi_m = \mathbb{E} D_m = 0$, this is equivalently the microcanonical covariance-deficit identity

$$(4.21) \quad \widehat{M}_m^{(\kappa, \tau)} = \begin{pmatrix} \frac{1}{2} \mathbb{E} \Sigma_m & 1 \\ 1 & \tau/g + \kappa m/2 \end{pmatrix} - \frac{1}{2} \text{Cov} \begin{pmatrix} \Xi_m \\ D_m \end{pmatrix}.$$

The formal structure “conditional expected curvature minus score covariance” in (4.21) is analogous to the missing-information identity for observed information in latent-variable models [27]; no identification with that likelihood framework is asserted here.

Finally, the normalized differential matrix is the canonical average of these fixed-total fibers:

$$(4.22) \quad \frac{1}{Z^2} \text{diag}(1, g^{-1/2}) \mathcal{T}_{\kappa, \tau} \text{diag}(1, g^{-1/2}) = \mathbb{E}_\lambda \widehat{M}_R^{(\kappa, \tau)}.$$

Proof. Independence and $[\lambda^m]Z^2 = S_m$ give the first identity in (4.18); division by that pair-total probability gives the conditional law. Under this law the u -score of a summand of $\mathcal{F}_m$ at the origin is $\Xi_m/\sqrt{2}$, its u -curvature is $-\Sigma_m/2$, and its v -score is $-D_m/\sqrt{2}$, while its v - and mixed curvatures vanish because v enters the summand linearly in the exponent and separately from u . Each second logarithmic derivative of a positive sum is the mean of the corresponding summandwise curvature plus the covariance of the two summandwise scores, and $\mathbb{E} \Xi_m = \mathbb{E} D_m = 0$ by symmetry, so

$$-\ell_{m,uu} = \frac{1}{2} \mathbb{E}(\Sigma_m - \Xi_m^2), \quad \ell_{m,uv} = -\frac{1}{2} \mathbb{E}(D_m \Xi_m), \quad \ell_{m,vv} = \frac{1}{2} \mathbb{E} D_m^2.$$

The identities in (4.20) are exactly (4.12), (4.14), and (4.15), so (4.19) follows from Theorem 4.2; (4.21) is the same identity grouped into baseline and covariance terms. Finally, weighting the coefficient expansion (4.8) by the pair-total law gives (4.22). $\square$

Corollary 4.6. *Let $\Psi = \psi(a + Y)$, $\mu = \mathbb{E} Y$, and $\sigma^2 = \text{Var } Y$. Then*

$$(4.23) \quad L_{\Theta_\lambda} = \mu, \quad L_{\Theta_\lambda \Theta_\lambda} = \sigma^2,$$

$$(4.24) \quad L_{a\Theta_\lambda} = -\text{Cov}(\Psi, Y), \quad -L_{aa} = \mathbb{E} \psi_1(a + Y) - \text{Var } \Psi.$$

Consequently (4.22) is the covariance-deficit matrix

$$(4.25) \quad \begin{pmatrix} \mathbb{E} \psi_1(a + Y) - \text{Var } \Psi & 1 - \text{Cov}(\Psi, Y) \\ 1 - \text{Cov}(\Psi, Y) & \tau/g + \kappa \mu - \sigma^2 \end{pmatrix}.$$

Equivalently,

$$(4.26) \quad \begin{pmatrix} \mathbb{E}\psi_1(a+Y) & 1 \\ 1 & \tau/g + \kappa\mu \end{pmatrix} - \text{Cov} \begin{pmatrix} \Psi \\ Y \end{pmatrix}$$

is the normalized matrix. Moreover

$$(4.27) \quad H_\nu^{(\kappa)} = 4(\kappa\mu - \sigma^2),$$

so $H_\nu > 0$ is exactly underdispersion of the Bessel law.

Proof. Average (4.20) over the independent pair (Y_1, Y_2) . Since

$$\begin{aligned} \frac{1}{2}\mathbb{E}\{\psi_1(a+Y_1) + \psi_1(a+Y_2)\} &= \mathbb{E}\psi_1(a+Y), \\ \frac{1}{2}\mathbb{E}(\Psi_2 - \Psi_1)^2 &= \text{Var } \Psi, \quad \frac{1}{2}\mathbb{E}\{(Y_2 - Y_1)(\Psi_2 - \Psi_1)\} = \text{Cov}(Y, \Psi), \end{aligned}$$

and

$$\frac{1}{2}\mathbb{E}(Y_2 - Y_1)^2 = \text{Var } Y,$$

comparison with Remark 2.2 proves the derivative and matrix identities. Equation (4.27) follows from (2.17). $\square$

At the endpoint, Theorem 2.5 is therefore equivalent to a covariance inequality for the discrete Bessel law (4.17):

$$(4.28) \quad (1 - \text{Cov}(\Psi, Y))^2 < (\mathbb{E}\psi_1(a+Y) - \text{Var } \Psi) \left(\frac{1}{g} + \mathbb{E}Y - \text{Var } Y \right).$$

4.2. Four-copy determinant sectors. For symmetric 2×2 matrices $U = (U_{ij})$ and $V = (V_{ij})$, define

$$\text{MD}(U, V) = U_{11}V_{22} + U_{22}V_{11} - 2U_{12}V_{12}.$$

This is the polarization of the determinant; up to the conventional factorial normalization, it is the mixed discriminant in dimension two [3].

Lemma 4.7. *If $U, V \succeq 0$, then $\text{MD}(U, V) \geq 0$. If in addition $U \neq 0$ and $V \succ 0$, then $\text{MD}(U, V) > 0$.*

Proof. Write $U = \sum_r u_r u_r^T$ and $V = \sum_s v_s v_s^T$. Then

$$\text{MD}(U, V) = \sum_{r,s} (u_{r,1}v_{s,2} - u_{r,2}v_{s,1})^2.$$

If this sum vanished and $U \neq 0$, every Gram vector of V would be parallel to a fixed nonzero Gram vector of U , forcing V to have rank at most one. $\square$

Let $R' = Y_3 + Y_4$ and $N = R + R'$. Define

$$(4.29) \quad T_n = \sum_{k=0}^n S_k S_{n-k} = [\lambda^n] Z^4, \quad \mathbb{P}(K_n = k) = \frac{S_k S_{n-k}}{T_n}.$$

Thus K_n has the conditional law of the first pair total R given $N = n$.

Proposition 4.8. *The canonical total has law*

$$(4.30) \quad \mathbb{P}_\lambda(N = n) = \frac{T_n \lambda^n}{Z^4}.$$

For every $n \geq 0$,

$$(4.31) \quad \Delta_n^{(\kappa, \tau)} = \frac{1}{2} \sum_{k=0}^n S_k S_{n-k} \text{MD} \left(M_k^{(\kappa, \tau)}, M_{n-k}^{(\kappa, \tau)} \right) = \frac{T_n}{2} \mathbb{E} \text{MD} \left(M_{K_n}^{(\kappa, \tau)}, M_{n-K_n}^{(\kappa, \tau)} \right).$$

If

$$(4.32) \quad d_n^{(\kappa, \tau)} = \frac{4\Delta_n^{(\kappa, \tau)}}{gT_n},$$

then the pointwise Bessel determinant is the canonical average of the microcanonical sector densities:

$$(4.33) \quad D_{a-1}^{(\kappa, \tau)}(2\sqrt{\lambda}) = \mathbb{E}_\lambda d_N^{(\kappa, \tau)}.$$

Proof. The total-count law follows from $T_n = [\lambda^n]Z^4$. Expanding the determinant of (4.8) by polarization gives

$$\Delta_n^{(\kappa, \tau)} = \frac{1}{2} \sum_{k=0}^n S_k S_{n-k} \text{MD}\left(M_k^{(\kappa, \tau)}, M_{n-k}^{(\kappa, \tau)}\right),$$

which is (4.31). Finally, combine (2.19) with (4.30) and (4.32). $\square$

Remark 4.9. Equation (4.33) makes the phase separation literal. Coefficientwise positivity asks for the sign of every positive fixed-total sector d_n , whereas pointwise positivity asks for the sign of their canonical mixture for every fugacity λ . The vacuum sector is $d_0^{(\kappa, \tau)} = 4(\tau - 1)$, so the pointwise correction wall is already visible before any positive-count sector is populated.

5. ENDPOINT COEFFICIENTWISE POSITIVITY

We require elementary two-sided trigamma estimates. The series $\psi_1(y) = \sum_{r \geq 0} (y + r)^{-2}$ is standard [33, eq. 5.15.1]; inserting $(y + r)^{-2} = \int_0^\infty \theta e^{-(y+r)\theta} d\theta$ and summing $\sum_{r \geq 0} e^{-r\theta} = (1 - e^{-\theta})^{-1}$ under the integral gives the integral representation

$$(5.1) \quad \psi_1(y) = \int_0^\infty e^{-y\theta} \frac{\theta}{1 - e^{-\theta}} d\theta, \quad y > 0.$$

Lemma 5.1. *For every $y > 0$,*

$$(5.2) \quad \psi_1(y) > \frac{1}{y} + \frac{1}{2y^2} > \frac{1}{y}.$$

For $y > 1/2$,

$$(5.3) \quad \psi_1(y) < \frac{1}{y - 1/2}.$$

For every $y > 0$,

$$(5.4) \quad \psi_1(y) < \frac{1}{y} + \frac{1}{2y^2} + \frac{1}{6y^3}.$$

Consequently,

$$(5.5) \quad \frac{1}{\psi_1(a)} > a - \frac{1}{2} \quad (a > 0),$$

where the assertion is immediate for $a \leq 1/2$.

Proof. For $h(s) = (y + s)^{-2}$, strict convexity on every interval $[r, r + 1]$ gives the strict trapezoidal inequality

$$\int_r^{r+1} h(s) ds < \frac{h(r) + h(r + 1)}{2}.$$

Summing and passing to the limit yields

$$\psi_1(y) = \sum_{r \geq 0} h(r) > \int_0^\infty h(s) ds + \frac{1}{2}h(0) = \frac{1}{y} + \frac{1}{2y^2},$$

which proves (5.2). For the first upper bound, the elementary inequality $\theta < 2 \sinh(\theta/2)$ for $\theta > 0$ is equivalent to $\theta/(1 - e^{-\theta}) < e^{\theta/2}$. Substitution in (5.1) gives (5.3). For the second, use

$$\frac{\theta}{1 - e^{-\theta}} = \frac{\theta}{2} \coth \frac{\theta}{2} + \frac{\theta}{2} < 1 + \frac{\theta}{2} + \frac{\theta^2}{12},$$

where $x \coth x < 1 + x^2/3$ for $x > 0$ because

$$(3 + x^2) \sinh x - 3x \cosh x = \sum_{k \geq 2} \frac{4k(k-1)}{(2k+1)!} x^{2k+1} > 0.$$

Integrating termwise in (5.1) gives (5.4). Taking reciprocals in (5.3) proves (5.5) when $a > 1/2$. $\square$

5.1. Gram structure and the exceptional matrix M_1 . Fix $m \geq 1$ and write $x = a + m$ and $q = a + m/2 - 1$. In $\ell^2(\mathbb{N}_0)$ define

$$u_r^{(m)} = \frac{1}{x+r}, \quad v_r^{(m)} = \frac{q}{x+r-1}, \quad r \geq 0.$$

Then

$$(5.6) \quad \|u^{(m)}\|^2 = \psi_1(x) = \alpha_m,$$

$$(5.7) \quad \langle u^{(m)}, v^{(m)} \rangle = q \sum_{r \geq 0} \frac{1}{(x+r-1)(x+r)} = \frac{q}{x-1} = \beta_m,$$

and

$$(5.8) \quad \|v^{(m)}\|^2 = q^2 \psi_1(x-1).$$

Define, for $s \in \mathbb{R}$,

$$(5.9) \quad \rho_m(s) = s + c_m - q^2 \psi_1(x-1),$$

and let $N_m(s)$ be the symmetric matrix with entries α_m, β_m and $s + c_m$.

For vectors ξ, η in a Hilbert space, $\text{Gram}(\xi, \eta)$ denotes the symmetric 2×2 matrix with entries $\langle \xi, \xi \rangle, \langle \xi, \eta \rangle, \langle \eta, \eta \rangle$. Write $N_m = N_m(1/g) = \text{diag}(1, g^{-1/2}) M_m \text{diag}(1, g^{-1/2})$ for the normalized coefficient matrix.

Theorem 5.2. *Let $m \geq 1$, $a > 0$ and $s \in \mathbb{R}$ with $\rho_m(s) > 0$. In $\ell^2(\mathbb{N}_0) \oplus \mathbb{R}$, put*

$$\xi_m = (u^{(m)}, 0), \quad \eta_m(s) = (v^{(m)}, \sqrt{\rho_m(s)}).$$

Then

$$(5.10) \quad N_m(s) = \text{Gram}(\xi_m, \eta_m(s))$$

is positive definite. If either $m \geq 2$ and $a > 0$, or $m = 1$ and $a \geq 1/2$, then $\rho_m(1/g) > 0$, so

$$(5.11) \quad M_m = \text{Gram}(\xi_m, \sqrt{g} \eta_m(1/g))$$

is positive definite in those ranges.

Proof. The Gram identity follows from (5.6)–(5.9): the final coordinate contributes $\rho_m(s)$, so the lower-right entry is $q^2 \psi_1(x-1) + \rho_m(s) = s + c_m$. In a vanishing combination $\mu \xi_m + \omega \eta_m(s) = 0$ the last coordinate gives $\omega \sqrt{\rho_m(s)} = 0$, hence $\omega = 0$, and then $\|u^{(m)}\|^2 = \psi_1(x) > 0$ forces $\mu = 0$. The two vectors are linearly independent, so $N_m(s) \succ 0$; the same combination with $\sqrt{g} \eta_m(1/g)$ gives (5.11).

It remains to verify $\rho_m(1/g) > 0$ in the stated ranges. Assume first that $m \geq 2$. Then $x - 1 > 1/2$, so Lemma 5.1 gives

$$\rho_m(1/g) > a - \frac{1}{2} + c_m - \frac{q^2}{x - 3/2}.$$

The right-hand side simplifies exactly to

$$a - \frac{1}{2} + \frac{m(m-1)}{2(2a+2m-3)} - \frac{(a+m/2-1)^2}{a+m-3/2} = \frac{m-1}{2(2a+2m-3)} > 0.$$

For $m = 1$, $q = a - 1/2$ and $c_1 = 0$, so $\rho_1(1/g) = 1/g - q^2 g$, which for $q \geq 0$ is positive exactly when $qg < 1$. At $a = 1/2$ one has $q = 0$ and the conclusion is immediate; for $a > 1/2$, $qg < 1$ is (5.3) at $y = a$. $\square$

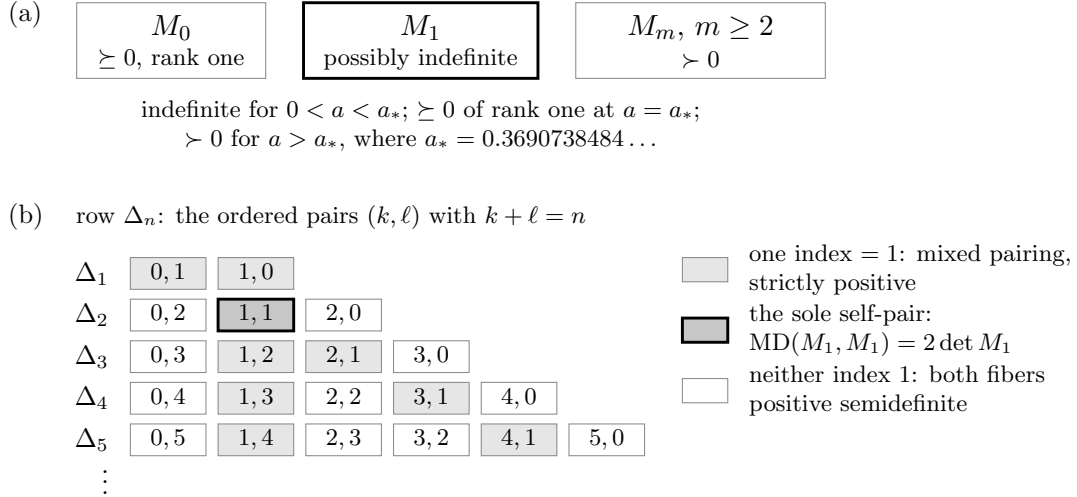


FIGURE 2. Finite-defect localization at the endpoint $(\kappa, \tau) = (1, 1)$. Panel (a): M_0 is positive semidefinite of rank one, $M_m \succ 0$ for $m \geq 2$ (Theorem 5.2), and M_1 is the only fiber whose inertia can change (Lemma 5.3). Panel (b): by (4.31), determinant degree n receives the ordered pairs (k, ℓ) with $k + \ell = n$, the pair (k, ℓ) contributing $S_k S_\ell \text{MD}(M_k, M_\ell)/2$. A pair with one index equal to 1 is a mixed pairing and is strictly positive, by (5.13) when the partner is M_0 and by (5.14) when it is M_m with $m \geq 2$; a pair with neither index equal to 1 has both fibers positive semidefinite. The sole self-pair $(1, 1)$ therefore occurs in Δ_2 and in no other degree, so the possible fiber-level defect is confined to one determinant degree, closed there by the explicit decomposition (5.17) (Lemma 5.5).

The constant matrix is

$$M_0 = \begin{pmatrix} g & \sqrt{g} \\ \sqrt{g} & 1 \end{pmatrix} = \begin{pmatrix} \sqrt{g} \\ 1 \end{pmatrix} \begin{pmatrix} \sqrt{g} & 1 \end{pmatrix},$$

a rank-one positive-semidefinite matrix. With $\psi_1(a+1) = g - a^{-2}$ and $\beta_1 = (2a-1)/(2a)$, the remaining coefficient matrix M_1 has determinant

$$(5.12) \quad \det M_1 = \psi_1(a+1) - g\beta_1^2 = \frac{(4a-1)g-4}{4a^2}.$$

Lemma 5.3. *There is $a_* = 0.3690738484\dots \in (0, 1/2)$ such that M_1 is indefinite for $0 < a < a_*$, positive semidefinite of rank one at $a = a_*$, and positive definite for $a > a_*$.*

Proof. Set $f(a) = (4a-1)\psi_1(a) - 4$, so that $\det M_1$ has the sign of $f(a)$ by (5.12). Since $g = \psi_1(a) = \sum_{r \geq 0} (a+r)^{-2}$, for $0 < a < 1/2$

$$f'(a) = 2 \sum_{r \geq 0} \frac{2r+1-2a}{(a+r)^3} > 0,$$

while $f(a) \rightarrow -\infty$ as $a \downarrow 0$ and $f(1/2) = \psi_1(1/2) - 4 = \pi^2/2 - 4 > 0$. Hence f has a unique zero $a_* \in (0, 1/2)$. The trichotomy follows with Theorem 5.2 for $a \geq 1/2$ and the positive diagonal of M_1 . $\square$

Among the coefficient matrices only M_1 can be indefinite at the endpoint $(\kappa, \tau) = (1, 1)$, so the determinant positivity of Theorem 2.3 cannot follow from pointwise semidefiniteness of the M_m alone; see Figure 2.

5.2. Finite-defect localization. The four-copy formula (4.31) makes the endpoint defect local in total degree. At $(\kappa, \tau) = (1, 1)$, the only coefficient fiber that can fail to be positive semidefinite is M_1 . Hence the bad self-pair (M_1, M_1) can occur only when the two pair totals

sum to 2. Every other determinant degree sees M_1 only through mixed pairings with a positive-definite fiber. More generally, a unique potentially indefinite fiber M_j in a 2×2 matrix series can self-interact under determinant polarization only in total degree $2j$.

Lemma 5.4. *For every $a > 0$ and every $n \geq 1$ with $n \neq 2$, one has $\Delta_n(a) > 0$.*

Proof. The linear coefficient is strictly positive: (5.2) gives $\psi_1(a) > 1/a$, so $ag > 1$ and $\text{MD}(M_0, M_1) = (ag - 1)/a^2 > 0$, whence

$$(5.13) \quad \Delta_1 = S_0 S_1 \text{MD}(M_0, M_1) = \frac{2(ag - 1)}{a^3 \Gamma(a)^4} > 0.$$

For $m \geq 2$ the mixed term satisfies

$$(5.14) \quad \text{MD}(M_1, M_m) = \alpha_1(1 + gc_m) + \alpha_m - 2g\beta_1\beta_m > 0.$$

If $a \geq 1/2$, then $M_1, M_m \succ 0$ by Theorem 5.2 and the strict case of Lemma 4.7 applies; if $0 < a < 1/2$, then $\beta_1 < 0 < \beta_m$ by (4.4), and since every diagonal entry is positive the right-hand side is strictly positive. For $n \geq 3$ the self-pair $(1, 1)$ does not occur. Classify the pairs $(k, n - k)$ in (4.31): if neither index is 1, both matrices in the pair are positive semidefinite— M_0 is rank-one positive semidefinite and $M_m \succ 0$ for $m \geq 2$ by Theorem 5.2—so the nonstrict case of Lemma 4.7 gives a nonnegative summand; if one index is 1, the other is at least 2 and (5.14) applies. Every summand is therefore nonnegative, and the pair (M_0, M_n) is strictly positive because $M_0 \neq 0$ and $M_n \succ 0$. Hence $\Delta_n > 0$. $\square$

5.3. The exceptional degree and endpoint sufficiency. The self-pairing $\text{MD}(M_1, M_1) = 2 \det M_1$ occurs solely in Δ_2 , so the indefiniteness established in Lemma 5.3 can contribute negatively only there.

Lemma 5.5. *For every $a > 0$, one has $\Delta_2(a) > 0$. More precisely, let*

$$t = \psi_1(a + 1), \quad \mathcal{R}_a = t - \frac{1}{a + 1}.$$

Then

$$(5.15) \quad \Delta_2 = \frac{Q_*(a, t)}{2a^6(a + 1)^3 \Gamma(a)^4},$$

where

$$(5.16) \quad \begin{aligned} Q_*(a, t) = & 2a^4(a + 1)^2 \mathcal{R}_a^2 \\ & + 2a^2(a + 1)^2(8a^2 + 3a + 1) \mathcal{R}_a \\ & + 2a(a + 1)(5a + 3). \end{aligned}$$

Proof. From (4.31),

$$(5.17) \quad \Delta_2 = S_0 S_2 \text{MD}(M_0, M_2) + S_1^2 \det M_1.$$

Here

$$S_0 = \frac{1}{\Gamma(a)^2}, \quad S_1 = \frac{2}{a\Gamma(a)^2}, \quad S_2 = \frac{2a + 1}{a^2(a + 1)\Gamma(a)^2},$$

and, recalling $\det M_1 = t - g\beta_1^2$ from (5.12),

$$\text{MD}(M_0, M_2) = g + \frac{g^2}{2a + 1} + \psi_1(a + 2) - 2g\frac{a}{a + 1}.$$

Substituting $g = t + a^{-2}$, $\psi_1(a + 2) = t - (a + 1)^{-2}$, and then $t = \mathcal{R}_a + (a + 1)^{-1}$ into (5.17), direct collection over the denominator in (5.15) gives the decomposition (5.16).

Applying (5.2) at $y = a + 1$ gives $\mathcal{R}_a = \psi_1(a + 1) - 1/(a + 1) > 1/(2(a + 1)^2) > 0$. All three coefficients in (5.16) are positive for $a > 0$, so $\Delta_2 > 0$. $\square$

Proof of Theorem 2.3, sufficiency. For every $a > 0$ and $n \geq 1$, the endpoint assertion $\Delta_n^{(1)}(a) > 0$ holds by Lemma 5.4 when $n \neq 2$ and by Lemma 5.5 when $n = 2$. Since A and Z have positive coefficients and

$$[\lambda^n]Z_{\Theta_\lambda} = \frac{n}{n!\Gamma(a+n)} > 0 \quad (n \geq 1),$$

every positive-degree coefficient of AZZ_{Θ_λ} is strictly positive: in degree n , for example, the product of the constant coefficients of A and Z with $[\lambda^n]Z_{\Theta_\lambda}$ is positive. Therefore

$$\Delta^{(\kappa)} = \Delta^{(1)} + (\kappa - 1)gAZZ_{\Theta_\lambda}$$

shows that all coefficients remain strictly positive for $\kappa > 1$.

Fix $\kappa \geq 1$. Since $\Delta^{(\kappa)}(a, \cdot)$ is entire, each differentiated Maclaurin series converges locally uniformly, so for every $r \geq 0$ and $\lambda > 0$,

$$\partial_\lambda^r \Delta^{(\kappa)}(a, \lambda) = \sum_{n \geq \max\{1, r\}} \frac{n!}{(n-r)!} \Delta_n^{(\kappa)}(a) \lambda^{n-r} > 0,$$

which makes $\Delta^{(\kappa)}(a, \cdot)$ strictly absolutely monotone. The leading principal entry satisfies $A > 0$ for $\lambda > 0$ by (4.2) and (4.3); since $\Delta^{(\kappa)} > 0$, it follows that $\mathcal{T}_\kappa \succ 0$. $\square$

The converse at each fixed $a > 0$ is deferred to the necessity part of Theorem 2.4.

Remark 5.6. Degree one already rules out any $\kappa < 1$ that is to serve every $a > 0$ at once. The general $m = 1$ coefficient on the slice $\tau = 1$ is

$$(5.18) \quad \text{MD}(M_0, M_1^{(\kappa, 1)}) = \frac{\frac{\kappa-1}{2}(ag)^2 + ag - 1}{a^2},$$

and if $\kappa < 1$, then $ag \rightarrow +\infty$ as $a \downarrow 0$, so (5.18), and hence $\Delta_1^{(\kappa)}$, is negative for all sufficiently small $a > 0$. This is weaker than the fixed- a necessity in Theorem 2.4, which rules out every $\kappa < 1$ at each fixed a .

Combining (5.13) with (5.2), which gives $ag - 1 > (2a)^{-1}$, yields the explicit lower bound

$$\Delta(a, \lambda) > \Delta_1(a)\lambda = \frac{2(a\psi_1(a) - 1)}{a^3\Gamma(a)^4} \lambda > \frac{\lambda}{a^4\Gamma(a)^4} \quad (a, \lambda > 0).$$

6. CANONICAL–MICROCANONICAL PHASE GEOMETRY

Proof of Theorem 2.5 and Corollary 2.6. The congruence (2.18) and the endpoint coefficient positivity established in Section 5.3 give positive definiteness for $z > 0$, and (2.21) gives the determinant inequality. As $z \downarrow 0$, equivalently $\lambda \downarrow 0$, the power series for $L = \log Z$ gives $L_{\Theta_\lambda}, L_{a\Theta_\lambda}, L_{\Theta_\lambda\Theta_\lambda} \rightarrow 0$, while $Z(a, 0) = 1/\Gamma(a)$ gives $L_{aa} \rightarrow -\psi_1(a)$. Hence (2.16)–(2.17) imply

$$G_\nu \longrightarrow g, \quad 1 + P_\nu \longrightarrow 2, \quad H_\nu + \frac{4}{g} \longrightarrow \frac{4}{g},$$

which is the matrix limit in Corollary 2.6. $\square$

6.1. Large argument and the deformation wall.

Lemma 6.1. *For fixed $\nu > -1$ and fixed $\kappa, \tau \in \mathbb{R}$,*

$$(6.1) \quad D_\nu^{(\kappa, \tau)}(z) = 2(\kappa - 1) + \frac{4\tau/g - 2(\kappa + 1)\nu - \frac{1}{2}}{z} + O(z^{-2}) \quad (z \rightarrow \infty).$$

The remainder is uniform when (κ, τ) ranges over a fixed compact subset of $\mathbb{R}^2$. In particular,

$$(6.2) \quad D_\nu(z) = \frac{c(a)}{z} + O(z^{-2}), \quad a = \nu + 1,$$

and $c(a) > 3/2$.

Proof. Differentiating a large-argument expansion of $\log I_\nu$ requires a zero-free complex neighborhood. Fix $\varphi \in (0, \pi/2)$, put $\Sigma_\varphi = \{w : |\arg w| \leq \pi/2 - \varphi\}$, and let $K \subset \mathbb{C}$ be a closed disc about the given order. Uniformly for $\mu \in K$ and $w \in \Sigma_\varphi$, the Hankel expansion and its error bounds in [32, eqs. 10.17.1 and 10.40.1, §10.40(iii)] give

$$(2\pi w)^{1/2} e^{-w} I_\mu(w) = 1 - \frac{4\mu^2 - 1}{8w} + \frac{(4\mu^2 - 1)(4\mu^2 - 9)}{128w^2} + O(w^{-3}).$$

For sufficiently large $|w|$, the left side lies in a fixed disc about 1, so its principal logarithm is holomorphic. Taking logarithms gives, uniformly on a slightly smaller product domain,

$$(6.3) \quad \log I_\mu(w) = w - \frac{1}{2} \log(2\pi w) - \frac{4\mu^2 - 1}{8w} - \frac{4\mu^2 - 1}{16w^2} + O(w^{-3}).$$

The remainder is holomorphic in (μ, w) on that domain, and for $\varepsilon = \varepsilon(\varphi)$ small the closed disc of radius $\varepsilon|w|$ about w lies in Σ_φ ; Cauchy estimates over that disc and over a fixed circle in the order variable therefore justify differentiating the remainder twice in μ and twice in w , each differentiation in w costing one power of $|w|$.

At $w = z > 0$, differentiation of (6.3) yields

$$G_\nu = \frac{1}{z} + \frac{1}{2z^2} + O(z^{-3}), \quad P_\nu = \frac{\nu}{z} + \frac{\nu}{z^2} + O(z^{-3}),$$

and

$$H_\nu^{(\kappa)} = (2\kappa - 1)z - \kappa(1 + 2\nu) + \frac{(2\kappa + 1)(4\nu^2 - 1)}{8z} + O(z^{-2}).$$

Substitution into (2.14) gives (6.1). Since κ and τ enter these expressions only affinely, the remainder is uniform on compact subsets of the (κ, τ) -plane. At $(\kappa, \tau) = (1, 1)$, its z^{-1} coefficient is $c(a)$. Finally (5.5) gives

$$c(a) = \frac{4}{g} - 4a + \frac{7}{2} > 4\left(a - \frac{1}{2}\right) - 4a + \frac{7}{2} = \frac{3}{2}.$$

On the other hand, (5.2) gives $g > 1/a$, hence $1/g < a$, so $c(a) < 7/2$. This proves (2.27). $\square$

6.2. The coefficient walls and the exposed degree-one boundary. We need one endpoint lemma for the lower boundary. Put

$$(6.4) \quad \tau_* = \tau_{\text{cw}}(a, 1) = \frac{ag(2a - 1)}{2a^2g - 1}, \quad s_* = \frac{\tau_*}{g} = \frac{a(2a - 1)}{2a^2g - 1}.$$

For the coefficient matrices on this boundary, normalize by

$$(6.5) \quad \hat{N}_m = \text{diag}(1, g^{-1/2}) M_m^{(1, \tau_*)} \text{diag}(1, g^{-1/2}) = \begin{pmatrix} \alpha_m & \beta_m \\ \beta_m & s_* + c_m \end{pmatrix}.$$

Lemma 6.2. *Let $a > 0$ and let $\hat{\Delta}_n = [\lambda^n] \Delta^{(1, \tau_*)}(a, \lambda)$. Then*

$$\hat{\Delta}_1 = 0, \quad \hat{\Delta}_n > 0 \quad (n \geq 2).$$

The proof is a four-part verification: positivity of the fibers with $m \geq 2$, positivity of mixed pairs involving the vacuum fiber, positivity of mixed pairs involving the one-particle fiber, and a separate check of the degree-two self-pair. The required trigamma estimates and algebraic reductions are collected in Appendix A.

We now write the coefficientwise region as the upper envelope of its individual degree walls. Put $p_n = [\lambda^n]P$ and $q_n = [\lambda^n]Q$. Since $ZZ_{\Theta_\lambda} = \frac{1}{2}\Theta_\lambda(Z^2)$, Theorem 4.2 gives

$$(6.6) \quad p_n = \sum_{k=0}^n S_k S_{n-k} \alpha_k, \quad q_n = \frac{g}{2} \sum_{k=0}^n (n-k) S_k S_{n-k} \alpha_k.$$

Recall the individual degree wall $\tau_n(a, \kappa)$ from (2.30). The affine determinant identity is exactly

$$(6.7) \quad [\lambda^n] \Delta^{(\kappa, \tau)} = p_n(\tau - \tau_n(a, \kappa)).$$

Thus positive-degree coefficientwise nonnegativity is the intersection of the half-planes $\tau \geq \tau_n(a, \kappa)$.

Under the symmetric law (4.29), $\mathbb{E}K_n = n/2$, and (6.6) gives the exact slope identity

$$(6.8) \quad \frac{q_n}{p_n} = \frac{gn}{4} - \frac{g}{2} \frac{\text{Cov}(K_n, \alpha_{K_n})}{\mathbb{E}\alpha_{K_n}}.$$

Since $k \mapsto \alpha_k = \psi_1(a + k)$ is strictly decreasing and the law of K_n is nondegenerate for $n \geq 1$, $\text{Cov}(K_n, \alpha_{K_n}) < 0$. Hence

$$(6.9) \quad \frac{q_n}{p_n} > \frac{gn}{4} \quad (n \geq 1).$$

For $n = 1$, direct calculation gives

$$(6.10) \quad \frac{q_1}{p_1} = \frac{a^2 g^2}{2(2a^2 g - 1)} < \frac{g}{2},$$

because $a^2 g > 1$. For $n \geq 2$, (6.9) gives $q_n/p_n > gn/4 \geq g/2 > q_1/p_1$, hence

$$(6.11) \quad \frac{q_n}{p_n} > \frac{q_1}{p_1} \quad (n \geq 2).$$

Equivalently, the higher-degree walls have strictly more negative κ -slope than the degree-one wall.

Proof of Theorem 2.4. At $\kappa = 1$, Lemma 6.2 says precisely that

$$\tau_1(a, 1) = \tau_*, \quad \tau_n(a, 1) < \tau_1(a, 1) \quad (n \geq 2).$$

Since $\partial_\kappa \tau_n = -q_n/p_n$, (6.11) gives, for $\kappa \geq 1$ and $n \geq 2$,

$$\tau_1(a, \kappa) - \tau_n(a, \kappa) = \tau_1(a, 1) - \tau_n(a, 1) + (\kappa - 1) \left(\frac{q_n}{p_n} - \frac{q_1}{p_1} \right) > 0.$$

Thus degree one is the unique exposed positive-degree wall and

$$\sup_{n \geq 1} \tau_n(a, \kappa) = \tau_1(a, \kappa) = \tau_* - (\kappa - 1) \frac{q_1}{p_1} = \tau_{\text{cw}}(a, \kappa).$$

Equation (6.7) now gives the strict and nonstrict coefficientwise assertions, including positivity of every degree $n \geq 2$ on the boundary.

It remains to exclude $\kappa < 1$. Fix τ . By Lemma 6.1, $D_\nu^{(\kappa, \tau)}(z) \rightarrow 2(\kappa - 1) < 0$. Since $Z(a, \lambda)^4 \rightarrow \infty$, (2.19) forces $\Delta^{(\kappa, \tau)}(a, \lambda) \rightarrow -\infty$. A power series whose positive-degree coefficients are all nonnegative is bounded below on $(0, \infty)$ by its constant coefficient, so some positive-degree coefficient must be negative.

Finally,

$$\tau_*(a) < 1 \iff a\psi_1(a) > 1,$$

which follows from (5.2); and $\tau_{\text{cw}}(a, \kappa)$ decreases with κ . This proves the last assertion as well. $\square$

Proof of Proposition 2.7. The endpoint theorem gives $D_\nu > 0$. From (2.14),

$$D_\nu^{(\kappa, \tau)} = D_\nu + 2(\kappa - 1)G_\nu(\Theta_z \log I_\nu - \nu) + \frac{4(\tau - 1)}{g}G_\nu.$$

For $\nu > -1, z > 0$, both $G_\nu = A/Z^2$ and $\Theta_z \log I_\nu - \nu = 2L_{\Theta_\lambda}$ are strictly positive. Thus $\kappa \geq 1, \tau \geq 1$ is sufficient.

As $z \downarrow 0$, the power series gives $D_\nu^{(\kappa, \tau)}(z) \rightarrow 4(\tau - 1)$, so pointwise positivity forces $\tau \geq 1$. As $z \rightarrow \infty$, Lemma 6.1 gives the limit $2(\kappa - 1)$, so it also forces $\kappa \geq 1$. This proves the sharp quadrant.

Now suppose $\kappa \geq 1$ and $\tau_{\text{cw}}(a, \kappa) \leq \tau < 1$. By Theorem 2.4, every positive-degree coefficient of $\Delta^{(\kappa, \tau)}$ is nonnegative and every coefficient of degree at least two is positive, while (2.8) is negative. Therefore

$$\frac{d}{d\lambda} \Delta^{(\kappa, \tau)}(a, \lambda) > 0 \quad (\lambda > 0),$$

and $\Delta^{(\kappa, \tau)}(a, \lambda) \rightarrow +\infty$ as $\lambda \rightarrow \infty$. It has exactly one positive zero. The positive factor in (2.19) transfers the same zero to $D_\nu^{(\kappa, \tau)}$. $\square$

Taking $\tau = 1$ in the necessity part of Theorem 2.4 supplies the deferred converse in Theorem 2.3; together with the endpoint sufficiency proved in Section 5.3, this completes that theorem.

7. CRITICAL WALL FAN AND EQUIVALENCE OF ENSEMBLES

The high-degree wall fan focuses at its thermodynamic accumulation point. There the coefficient–argument correspondence becomes an instance of equivalence between the canonical and microcanonical ensembles.

7.1. The central microcanonical law. With the four-copy conditional variable K_n from (4.29), put $X_n = (K_n - n/2)/\sqrt{n}$. Thus X_n is the centered first-pair total under the microcanonical constraint $Y_1 + Y_2 + Y_3 + Y_4 = n$. By (4.31) and (6.6), symmetry gives

$$(7.1) \quad n(1 - \kappa_n) = \frac{n \mathbb{E} \text{MD}(M_{K_n}, M_{n-K_n})}{g \mathbb{E}[(n - K_n)\alpha_{K_n}]}.$$

The law of K_n is a slowly varying tilt of a symmetric hypergeometric law. We record a quantitative form because it is what makes the remainder in Theorem 2.8 effective.

Lemma 7.1. *For fixed $a > 0$,*

$$(7.2) \quad \mathbb{E}X_n^2 = \frac{1}{8} + \frac{2a-1}{8n} + O_a(n^{-2}),$$

$$(7.3) \quad \mathbb{E}X_n^4 = \frac{3}{64} + O_a(n^{-1}), \quad \mathbb{E}|X_n|^6 = O_a(1).$$

More strongly, for every fixed $q, M \geq 0$,

$$(7.4) \quad \mathbb{E}\left[(1 + |X_n|^q)\mathbf{1}_{\{|K_n - n/2| > n^{3/5}\}}\right] = O_{a,q,M}(n^{-M}).$$

All these estimates are uniform when a ranges over a compact subinterval of $(0, \infty)$.

Proof. The duplication formula rewrites S_m , for all sufficiently large m , as

$$(7.5) \quad S_m = \frac{2^{2a-2}}{\sqrt{\pi}} \frac{4^m}{(m!)^2} \frac{\Gamma(m+a-1/2)}{\Gamma(m+a)} \frac{\Gamma(m+1)}{\Gamma(m+2a-1)}.$$

The standard gamma-ratio expansion [33, §5.11(iii)] gives

$$(7.6) \quad S_m = C_a \frac{4^m}{(m!)^2} m^b \left(1 + \frac{s_a}{m} + O_a(m^{-2})\right),$$

where

$$C_a = \frac{2^{2a-2}}{\sqrt{\pi}}, \quad b = \frac{3}{2} - 2a, \quad s_a = -2a^2 + \frac{5}{2}a - \frac{5}{8}.$$

If the slowly varying factors are omitted, (4.29) becomes

$$\pi_{n,k} = \frac{\binom{n}{k}^2}{\binom{2n}{n}},$$

the hypergeometric law obtained by drawing n points without replacement from a population of size $2n$ containing n marked points.

For quantitative tails, set

$$h_m(a) = \frac{(m!)^2}{4^m} S_m(a), \quad u_m(a) = \frac{h_m(a)}{(m+1)^b}.$$

By (7.6), $u_m(a) \rightarrow C_a > 0$. Hence, for fixed a ,

$$0 < m_a := \inf_{m \geq 0} u_m(a) \leq \sup_{m \geq 0} u_m(a) =: M_a < \infty.$$

The exact law of K_n is

$$\mathbb{P}(K_n = k) = \frac{\pi_{n,k} h_k h_{n-k}}{\mathbb{E}_{\pi_n}[h_K h_{n-K}]}.$$

On the event $|K - n/2| \leq \sqrt{n}$, whose π_n -probability is bounded away from zero by the exact variance below and Chebyshev's inequality, the factor $h_K h_{n-K}$ is bounded below by a positive multiple of n^{2b} , because $k+1$ and $n-k+1$ are both of order n there. Globally $h_k h_{n-k} \leq M_a^2[(k+1)(n-k+1)]^b$, and on $0 \leq k \leq n$ the product $(k+1)(n-k+1)$ lies in $[n+1, (n/2+1)^2]$, so that bound is largest near the center when $b \geq 0$ and at the endpoints when $b < 0$; in both cases it is a positive multiple of $n^{2b+(-b)+}$. Therefore there is a finite constant $C_{\text{tilt}}(a)$ such that, for every event A ,

$$(7.7) \quad \mathbb{P}(K_n \in A) \leq C_{\text{tilt}}(a) n^{(-b)+} \pi_n(A).$$

The constants may be chosen uniformly for a in a compact subinterval of $(0, \infty)$.

Hoeffding's sampling-without-replacement inequality [18, §6] gives

$$\pi_n\{|K - n/2| > t\} \leq 2e^{-2t^2/n}.$$

With $t = n^{3/5}$, (7.7) yields

$$\mathbb{P}\{|K_n - n/2| > n^{3/5}\} \leq 2C_{\text{tilt}}(a) n^{(-b)+} e^{-2n^{1/5}},$$

which implies (7.4) because $|X_n| \leq \sqrt{n}/2$.

Under the base law π_n , for $X = (K - n/2)/\sqrt{n}$, exact factorial moments give

$$(7.8) \quad \mathbb{E}_0 X^2 = \frac{n}{4(2n-1)} = \frac{1}{8} + \frac{1}{16n} + O(n^{-2}),$$

$$\mathbb{E}_0 X^4 = \frac{n(3n-4)}{16(2n-3)(2n-1)} = \frac{3}{64} + O(n^{-1}), \quad \mathbb{E}_0 |X|^6 = O(1).$$

On the central window $|K - n/2| \leq n^{3/5}$, one has $|x| \leq n^{1/10} = o(n^{1/6})$ and

$$[k(n-k)]^b = \left(\frac{n^2}{4}\right)^b \left(1 - \frac{4bx^2}{n} + O_a\left(\frac{1+x^4}{n^2}\right)\right).$$

The product of the s_a/m corrections in (7.6) is $1 + 4s_a/n + O_a((1+x^2)/n^2)$, and its order-1/ n constant cancels under normalization. The super-polynomial tail estimate permits termwise averaging, so

$$\mathbb{E} X_n^2 = \mathbb{E}_0 X^2 - \frac{4b}{n} (\mathbb{E}_0 X^4 - (\mathbb{E}_0 X^2)^2) + O_a(n^{-2}).$$

Substitution proves (7.2); the fourth- and sixth-moment claims follow in the same way. $\square$

7.2. Asymptotics of the degree thresholds. The exact formulas in Theorem 4.2 have the large- m expansions

$$(7.9) \quad \alpha_m = \frac{1}{m} + \frac{1/2 - a}{m^2} + \frac{a^2 - a + 1/6}{m^3} + O(m^{-4}),$$

$$(7.10) \quad \beta_m = \frac{1}{2} + \frac{a-1}{2m} - \frac{(a-1)^2}{2m^2} + O(m^{-3}),$$

$$(7.11) \quad c_m = \frac{m}{4} - \frac{a-1/2}{4} + \frac{(a-3/2)(a-1/2)}{4m} + O(m^{-2}).$$

For $k = n/2 + x\sqrt{n}$ and $\ell = n - k$, substitution in the mixed determinant gives, uniformly on any growing central window on which $|x| = o(n^{1/6})$,

$$(7.12) \quad \text{MD}(M_k, M_\ell) = \frac{\Phi_1(x)}{n} + \frac{\Phi_2(x)}{n^2} + O\left(\frac{1+|x|^6}{n^3}\right),$$

where

$$(7.13) \quad \Phi_1(x) = 4 + g(3 - 4a) + 4gx^2$$

and

$$(7.14) \quad \begin{aligned} \Phi_2(x) = & \frac{1}{3}(48gx^4 + (60 - 96a)gx^2 + 48x^2 \\ & + 24a^2g - 36ag - 24a + 13g + 12). \end{aligned}$$

Likewise, symmetry gives

$$(7.15) \quad \frac{1}{2}[\ell\alpha_k + k\alpha_\ell] = 1 + \frac{1 - 2a + 8x^2}{n} + O\left(\frac{1 + x^4}{n^2}\right).$$

The tail estimates in Lemma 7.1 make these expansions integrable under the law of K_n . The large- m expansions above and the central-window remainders are uniform when a ranges over a compact subinterval of $(0, \infty)$. On the central window, the remainders in (7.12) and (7.15) have expectations $O_a(n^{-3})$ and $O_a(n^{-2})$, respectively, by (7.3). Off that window, uniformly for a in such compact sets, the exact coefficient formulas give $\text{MD}(M_k, M_{n-k}) = O(n)$ and $(n-k)\alpha_k = O(n)$, with uniform implied constants; choosing M sufficiently large in (7.4) makes those tail contributions smaller than the displayed remainders.

Taking expectations and using (7.2)–(7.3) gives

$$(7.16) \quad \mathbb{E} \text{MD}(M_{K_n}, M_{n-K_n}) = \frac{gc(a)}{n} + \frac{96a^2g - 180ag - 96a + 85g + 72}{12n^2} + O_a(n^{-3}),$$

and

$$(7.17) \quad \mathbb{E}[(n - K_n)\alpha_{K_n}] = 1 + \frac{2(1 - a)}{n} + O_a(n^{-2}).$$

More explicitly,

$$\mathbb{E}\Phi_1(X_n) = gc(a) + \frac{g(2a - 1)}{2n} + O_a(n^{-2}),$$

while

$$\mathbb{E}\Phi_2(X_n) = \frac{96a^2g - 192ag - 96a + 91g + 72}{12} + O_a(n^{-1}).$$

Combining these two contributions gives the second coefficient in (7.16). At leading order the $4gx^2$ term in (7.13) contributes $4g \cdot (1/8) = g/2$; this Gaussian fluctuation is the source of the additional $1/2$ in the critical constant.

Proof of Theorem 2.8. Insert (7.16) and (7.17) into (7.1). Division gives

$$n(1 - \kappa_n) = c(a) + \frac{g - 24}{12g} \frac{1}{n} + O_a(n^{-2}),$$

which is equivalent to (2.25). The first equality in (2.26) follows from (6.2), and the second from the leading term just proved. The bounds in (2.27) were proved in Lemma 6.1; strict monotonicity and the endpoint limits of $c(a)$ are proved in Proposition 7.3 below.

Since $c(a) > 3/2$,

$$\kappa_{n+1} - \kappa_n = \frac{c(a)}{n^2} + O_a(n^{-3}) > 0$$

for all sufficiently large n . Finally, $\Delta_n^{(\kappa)} = q_n(\kappa - \kappa_n)$ with $q_n > 0$. Since the endpoint coefficients satisfy $\Delta_n > 0$, one has $\kappa_n < 1$ for every n . Thus for each fixed J , $\max_{1 \leq n \leq J} \kappa_n < 1$, so $N_a(\kappa) > J$ once κ is sufficiently close to 1; in particular, $N_a(\kappa) \rightarrow \infty$ as $\kappa \uparrow 1$. The first negative coefficient therefore lies in the eventually monotone tail. Since $1 - \kappa_n = c(a)/n + O_a(n^{-2})$, it is the first index for which $1 - \kappa_n < 1 - \kappa$, and inversion gives (2.28). The two-term expansion in fact yields $N_a(\kappa) = c(a)/(1 - \kappa) + O(1)$. $\square$

The critical-scaling theorem strengthens the fixed- a necessity in Theorem 2.4: below the deformation wall, negativity is not merely present in some degree but eventually persists throughout the coefficient tail.

Corollary 7.2. *Fix $a > 0$ and $\kappa < 1$. Then $\Delta_n^{(\kappa)}(a) < 0$ for every sufficiently large n .*

Proof. By (2.25), $\kappa_n(a) \rightarrow 1$. Since $\Delta_n^{(\kappa)} = q_n(\kappa - \kappa_n)$ and $q_n > 0$, one has $\kappa < \kappa_n$ for all sufficiently large n . $\square$

7.3. The coefficient-wall fan. The same central law gives the geometry of every fixed- τ deformation threshold; Figure 3 shows the resulting fan at $a = 1$. From (6.6) and (4.29),

$$\frac{p_n}{T_n} = \mathbb{E}\alpha_{K_n}, \quad \frac{q_n}{T_n} = \frac{g}{2} \mathbb{E}[(n - K_n)\alpha_{K_n}].$$

Using (7.9), symmetry, (7.2), and the tail estimate (7.4), termwise averaging gives

$$(7.18) \quad \frac{p_n}{T_n} = \frac{2}{n} + \frac{3 - 4a}{n^2} + O_a(n^{-3}).$$

Together with (7.17), this yields the sharper ratio

$$(7.19) \quad \frac{p_n}{q_n} = \frac{4}{gn} - \frac{2}{gn^2} + O_a(n^{-3}).$$

Proof of Theorem 2.9. By (2.29),

$$1 - \kappa_{n,\tau} = (1 - \kappa_n) + (\tau - 1) \frac{p_n}{q_n}.$$

Substitution of (2.25) and (7.19) gives (2.32) with c_τ, d_τ as in (2.31). The compact-uniform versions of the same estimates give local uniformity in (a, τ) .

For the Bessel side, extend (6.3) by one term. Uniformly for the order in a compact set and z in a fixed proper right-half-plane sector,

$$(7.20) \quad \begin{aligned} \log I_\nu(z) = z - \frac{1}{2} \log(2\pi z) - \frac{4\nu^2 - 1}{8z} - \frac{4\nu^2 - 1}{16z^2} \\ + \frac{(4\nu^2 - 1)(4\nu^2 - 25)}{384z^3} + O(z^{-4}). \end{aligned}$$

The same Cauchy-estimate argument used in Lemma 6.1 permits the required order and argument derivatives. Substitution in (2.14) at $\kappa = 1$ gives

$$D_{a-1}^{(1,\tau)}(z) = \frac{4\tau/g - 4a + 7/2}{z} + \frac{2\tau/g - 3(a - 1) - 1/3}{z^2} + O_{a,\tau}(z^{-3}),$$

which is (2.33).

The first coefficient c_τ vanishes precisely at $\tau = \tau_\infty(a) = g(a - 7/8)$. At this value,

$$d_{\tau_\infty}(a) = 2 \left(\frac{11}{12} - a \right), \quad e_{\tau_\infty}(a) = \frac{11}{12} - a,$$

which proves (2.35).

The coefficient walls of (2.30) satisfy, at $\kappa = 1$,

$$\tau_n(a, 1) = 1 - \frac{\Delta_n}{p_n}.$$

Since $\Delta_n/q_n = 1 - \kappa_n$, (2.25) and (7.19) give

$$(7.21) \quad \lim_{n \rightarrow \infty} \tau_n(a, 1) = 1 - \frac{gc(a)}{4} = g \left(a - \frac{7}{8} \right) = \tau_\infty(a).$$

Thus τ_∞ is the thermodynamic accumulation point of the positive-degree wall fan.

The inequality $\tau_{\text{cw}}(a, 1) < 1$ was proved in Theorem 2.4. For the other side, direct subtraction gives

$$\tau_{\text{cw}}(a, 1) - \tau_\infty(a) = -\frac{g\{16a^3g - 14a^2g - 16a^2 + 7\}}{8(2a^2g - 1)}.$$

If $a \leq 7/8$, the lower bound in (5.2) makes the expression in braces at most $-6a < 0$, with strict inequality for $a < 7/8$. If $a \geq 7/8$, the upper bound (5.4) makes it strictly smaller than $-(18a^2 - 8a + 7)/(3a) < 0$. Hence $\tau_\infty < \tau_{\text{cw}}(a, 1)$, completing (2.36). $\square$

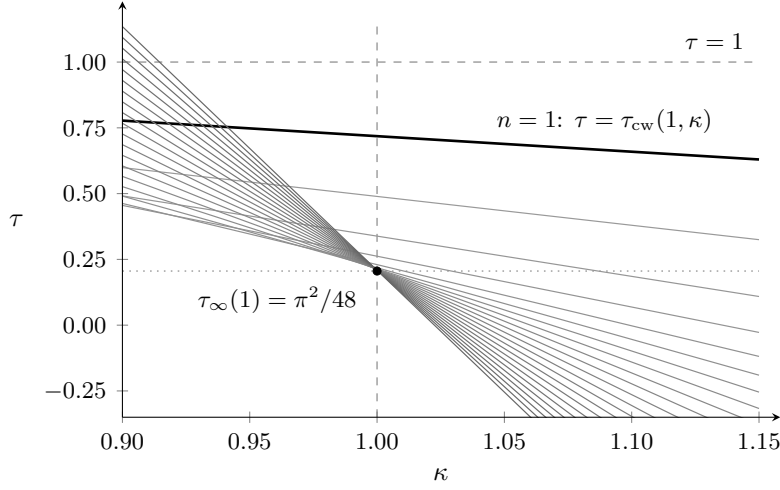


FIGURE 3. Coefficient-wall fan at $a = 1$. The lines are the exact degree walls $\tau = \tau_n(1, \kappa)$ of (2.30), $1 \leq n \leq 22$. Their high-degree members focus at the thermodynamic accumulation point $(1, \tau_\infty(1))$, $\tau_\infty(1) = \pi^2/48$, while the $n = 1$ wall is the exposed lower boundary $\tau = \tau_{cw}(1, \kappa)$ for nonnegative positive-degree coefficients when $\kappa \geq 1$ (Theorem 2.4). Since $\partial_\kappa \tau_n = -q_n/p_n \sim -\psi_1(1)n/4$, the high-degree walls rise without bound at fixed $\kappa < 1$ and fall without bound at fixed $\kappa > 1$; the former is the geometry behind the necessity of $\kappa \geq 1$. The value $a = 1$ is representative; the asymptotic statements hold for every $a > 0$ (Theorem 2.9 and Corollary 2.11).

Proposition 7.3. *The function $c(a) = 4/\psi_1(a) - 4a + 7/2$ is strictly decreasing on $(0, \infty)$, with*

$$\lim_{a \downarrow 0} c(a) = \frac{7}{2}, \quad \lim_{a \rightarrow \infty} c(a) = \frac{3}{2}.$$

Thus $(3/2, 7/2)$ is its exact range.

Proof. The inequality $\psi_1(a)^2 + \psi_2(a) > 0$ used below is classical in the polygamma-inequality literature; see Alzer [1, eq. (4.39) in the proof of Thm. 4.8]. We keep the following short derivation to make the monotonicity argument self-contained. Write

$$g(a) = \psi_1(a) = \int_0^\infty e^{-at} k(t) dt, \quad k(t) = \frac{t}{1 - e^{-t}}.$$

Then

$$g(a)^2 = \int_0^\infty e^{-as} (k * k)(s) ds, \quad -g'(a) = \int_0^\infty e^{-as} \frac{s^2}{1 - e^{-s}} ds.$$

For $0 < t < s$,

$$\frac{1 - e^{-s}}{(1 - e^{-t})(1 - e^{-(s-t)})} = \frac{1}{1 - e^{-t}} + \frac{1}{1 - e^{-(s-t)}} - 1.$$

Using $t/(1 - e^{-t}) = \frac{t}{2} \coth \frac{t}{2} + \frac{t}{2} > 1 + \frac{t}{2}$, valid because $x \coth x > 1$ for $x > 0$ (indeed, $x \cosh x - \sinh x$ vanishes at 0 and has derivative $x \sinh x > 0$), and symmetry,

$$\begin{aligned} (1 - e^{-s})(k * k)(s) &= 2 \int_0^s \frac{t(s-t)}{1 - e^{-t}} dt - \int_0^s t(s-t) dt \\ &> 2 \int_0^s (s-t) \left(1 + \frac{t}{2}\right) dt - \frac{s^3}{6} = s^2. \end{aligned}$$

Hence $g^2 + g' > 0$, and therefore

$$c'(a) = -4 \frac{g^2 + g'}{g^2} < 0.$$

As $a \downarrow 0$, the partial-fraction series gives $g(a) = a^{-2} + O(1)$, so $c(a) \rightarrow 7/2$. As $a \rightarrow \infty$, Lemma 5.1 gives

$$g(a) = \frac{1}{a} + \frac{1}{2a^2} + O(a^{-3}),$$

hence $1/g = a - 1/2 + O(a^{-1})$ and $c(a) \rightarrow 3/2$. $\square$

7.4. Equivalence of ensembles and the coefficient–argument dictionary. The exact identity (4.33) already places the pointwise and coefficient problems in one ensemble. Conditioning exponential-family samples on sufficient statistics and comparing the conditioned laws with tilted product laws has a classical general theory [13]; for the broader canonical–microcanonical ensemble-equivalence terminology see, for example, [42]. From the generating-function side, the use of a parameterized random-size law and its conditioning or tuning to fixed size is the classical Boltzmann paradigm [14]; recent work also develops fixed-size/random-size transfer for moment statistics [31]. The present argument is more explicit: the finite Bessel sector decomposition is exact, its sectors carry the Schur-deficit and mixed-determinant structure, and the remaining scale dictionary is concentration of the four-copy canonical total. From (4.23) and (2.16),

$$(7.22) \quad \mathbb{E}_\lambda N = 4L_{\Theta_\lambda} = 2(\Theta_z \log I_{a-1}(z) - (a-1)) = 2z - (2a-1) + O_a(z^{-1}).$$

Moreover, (4.27) at $\kappa = 1$ gives

$$\text{Var } N = 4 \text{Var } Y < 4\mathbb{E}Y = \mathbb{E}N = O_a(z).$$

Therefore

$$(7.23) \quad \frac{N}{2z} \longrightarrow 1 \quad (z \rightarrow \infty)$$

in probability. In particular, the microcanonical total corresponding to the canonical argument z is $n \sim 2z$.

The usual coefficient saddle is the analytic form of the same identity: for Z^4 , the radial stationarity equation is

$$n = \lambda \partial_\lambda \log Z^4 = \mathbb{E}_\lambda N.$$

Thus the factor 2 in the coefficient–argument correspondence is not an independent saddle coincidence; it is equivalence of the canonical and microcanonical descriptions.

For example, when $\tau = 1$ and $\kappa = 1 - \varepsilon$, Lemma 6.1 gives, uniformly for small $\varepsilon \geq 0$,

$$D_{a-1}^{(1-\varepsilon,1)}(z) = -2\varepsilon + \frac{c(a) + 2\varepsilon(a-1)}{z} + O_a(z^{-2}).$$

The sign-bracketing argument with $z = s/\varepsilon$ gives a positive zero z_ε satisfying $z_\varepsilon \sim c(a)/(2\varepsilon)$. Together with (2.28), $N_a(1-\varepsilon) \sim 2z_\varepsilon$, exactly as predicted by (7.23).

At fixed (κ, τ) , the sector densities themselves satisfy

$$d_n^{(\kappa,\tau)} = \frac{4q_n}{gT_n}(\kappa - \kappa_{n,\tau}),$$

so (7.17) and (2.32) give

$$\lim_{n \rightarrow \infty} d_n^{(\kappa,\tau)} = 2(\kappa - 1).$$

The three phase walls are therefore literally the contacts of the sector family at $n = 0$, $n = 1$, and $n = \infty$.

7.5. The cubic multicritical point.

Proof of Theorem 2.10. Recall (2.37). At this value the gamma-ratio expansion (7.5) can be continued to

$$(7.24) \quad S_m = C_* \frac{4^m}{(m!)^2} m^{-1/3} \left(1 - \frac{1}{72m} - \frac{169}{10368m^2} + O(m^{-3}) \right), \quad C_* = C_{11/12} > 0.$$

On the central scale $k = n/2 + x\sqrt{n}$, $\ell = n - k$, the slowly varying tilt relative to the symmetric hypergeometric law is therefore

$$(7.25) \quad 1 + \frac{24x^2 - 1}{18n} + \frac{192x^4 - 16x^2 - 7}{54n^2} + O\left(\frac{1 + |x|^6}{n^3}\right),$$

up to a factor independent of x . The base hypergeometric moments satisfy

$$\begin{aligned} \mathbb{E}_0 X^2 &= \frac{n}{4(2n-1)}, & \mathbb{E}_0 X^4 &= \frac{n(3n-4)}{16(2n-3)(2n-1)}, \\ \mathbb{E}_0 X^6 &= \frac{15n^3 - 60n^2 + 64n - 16}{64(2n-5)(2n-3)(2n-1)}. \end{aligned}$$

Normalizing (7.25) and averaging these exact moments gives

$$(7.26) \quad \begin{aligned} \mathbb{E} X_n^2 &= \frac{1}{8} + \frac{5}{48n} + \frac{35}{288n^2} + O(n^{-3}), \\ \mathbb{E} X_n^4 &= \frac{3}{64} + \frac{1}{16n} + O(n^{-2}), & \mathbb{E} X_n^6 &= \frac{15}{512} + O(n^{-1}), & \mathbb{E} X_n^8 &= O(1). \end{aligned}$$

The tail estimate (7.4) remains available with arbitrary polynomial weight.

Define

$$\mathcal{H}_{n,k} = \frac{1}{2} \text{MD}(M_k, M_\ell) + (\tau_{\text{mc}} - 1) \frac{\alpha_k + \alpha_\ell}{2}.$$

Expanding the exact coefficient formulas one order beyond (7.12) gives, uniformly on $|x| = o(n^{1/8})$,

$$(7.27) \quad \begin{aligned} \mathcal{H}_{n,k} &= \frac{g_{\text{mc}}(8x^2 - 1)}{4n} + \frac{g_{\text{mc}}(192x^4 - 104x^2 - 1)}{24n^2} \\ &+ \frac{g_{\text{mc}}(1536x^6 - 1472x^4 + 176x^2 - 3)}{48n^3} + O\left(\frac{1 + |x|^8}{n^4}\right). \end{aligned}$$

Substitution of (7.26) cancels the n^{-1} and n^{-2} expectations and yields

$$(7.28) \quad \mathbb{E} \mathcal{H}_{n,K_n} = \frac{3g_{\text{mc}}}{16n^3} + O(n^{-4}).$$

But

$$\frac{\Delta_n + (\tau_{\text{mc}} - 1)p_n}{T_n} = \mathbb{E} \mathcal{H}_{n,K_n}, \quad \frac{q_n}{T_n} = \frac{g_{\text{mc}}}{2} (1 + O(n^{-1})),$$

so (2.29) gives (2.38).

For the Bessel side, continue the logarithmic Hankel expansion by one further term. With $\chi = 4\nu^2$,

$$(7.29) \quad \begin{aligned} \log I_\nu(z) &= z - \frac{1}{2} \log(2\pi z) - \frac{\chi - 1}{8z} - \frac{\chi - 1}{16z^2} + \frac{(\chi - 1)(\chi - 25)}{384z^3} \\ &+ \frac{(\chi - 1)(\chi - 13)}{128z^4} + O(z^{-5}). \end{aligned}$$

The Cauchy-estimate argument used in Lemma 6.1 permits differentiation of the remainder. At $\nu = -1/12$, $\tau = g_{\text{mc}}/24$, substitution in (2.14) gives

$$D_{-1/12}^{(1, \tau_{\text{mc}})}(z) = \frac{3}{32z^3} + O(z^{-4}),$$

which is (2.39). □

Proof of Corollary 2.11. At $\kappa = 1$,

$$\tau_n(a, 1) = 1 - \frac{\Delta_n/q_n}{p_n/q_n}.$$

Dividing (2.25) by (7.19) gives (2.40). The stated one-sided approach to τ_∞ follows immediately. At $a = a_{\text{mc}}$, the same wall line satisfies

$$p_n\{\tau_{\text{mc}} - \tau_n(a_{\text{mc}}, 1)\} = q_n\{1 - \kappa_{n, \tau_{\text{mc}}}(a_{\text{mc}})\}.$$

Using (2.38) and $q_n/p_n = g_{\text{mc}}n/4 + O(1)$, the reciprocal of (7.19), gives (2.41).

For general κ , the affine wall equation gives

$$\tau_n(a, \kappa) = \tau_n(a, 1) - (\kappa - 1) \frac{q_n}{p_n}.$$

Since $q_n/p_n = gn/4 + O_a(1)$, this proves (2.42); taking $\kappa = 1 - x/n$ gives (2.43). $\square$

Remark 7.4. The first three verified wall jets obey one transfer rule. If a microcanonical threshold has leading form $1 - \kappa_{n,\tau} \sim A_r^{\text{mic}} n^{-r}$ and the corresponding canonical Bessel wall has $D_{a-1}^{(1,\tau)}(z) \sim A_r^{\text{can}} z^{-r}$, then the three cases $r = 1, 2, 3$ proved above satisfy $A_r^{\text{can}} = 2^{1-r} A_r^{\text{mic}}$. Indeed, $n \sim 2z$ and the pointwise κ -deformation enters as $-2(1 - \kappa)$, so this factor is exactly what canonical–microcanonical matching predicts. The successive instances are

$$c_\tau \leftrightarrow c_\tau, \quad 2(11/12 - a) \leftrightarrow 11/12 - a, \quad \frac{3}{8} \leftrightarrow \frac{3}{32}.$$

8. RELATION TO EXISTING POSITIVITY THEORIES

The results combine several classical one-dimensional positivity mechanisms but are not reducible to any one of them.

8.1. Reciprocal-gamma and hypergeometric Turánians. The scalar quantity $A = Z_a^2 - ZZ_{aa}$ is the infinitesimal symmetric-shift Turánian of the reciprocal-gamma series:

$$Z(a + \delta, \lambda)Z(a - \delta, \lambda) - Z(a, \lambda)^2 = -A(a, \lambda)\delta^2 + O(\delta^4).$$

Its coefficient sign is the infinitesimal form of the reciprocal-gamma Turán inequalities of Kalmykov and Karp [21, Thm. 3.1 and Ex. 1]; related gamma-ratio and hypergeometric inequalities appear in [22, 23, 24, 25, 26]. Those are scalar parameter-shift inequalities. Here the new object is the coupled determinant $AC - \psi_1(a)B^2$, and the two deformation parameters separate an affine coefficientwise boundary from a distinct pointwise boundary. The intermediate region in which the determinant has exactly one positive zero, and the high-degree critical law, have no scalar parameter-shift analogue in these results.

8.2. Bessel ratios, Turán inequalities, and order derivatives. By Proposition 3.1,

$$H_\nu = z^2 \frac{I_{\nu+1}^2 - I_\nu I_{\nu+2}}{I_\nu^2} > 0$$

is exactly a classical shifted modified-Bessel Turán inequality, equivalently the Amos-type lower ratio bound (3.5). Amos established sharp ratio bounds and monotonicity properties for non-negative order [2]; Näsell developed convergent rational bounds on ranges extending below zero [30]; Segura linked sharp ratio bounds to Turán inequalities and later classified best simple algebraic bounds [39, 40]; Hornik and Grün gave a systematic classification of Amos-type bounds on the full positivity range $\nu \geq -1$ [19]. Their Theorem 3 characterizes the lower Amos-type bounds $t/(\alpha + \sqrt{t^2 + \beta^2})$ for R_ν as those with $\alpha \geq \nu + \frac{1}{2}$, $\alpha + \beta \geq 2(\nu + 1)$ and $\beta \geq 0$; the diagonal $\alpha = \beta = \nu + \kappa$ of that family is admissible exactly when $\kappa \geq 1$, which is (3.6). The present paper does not improve these scalar ratio bounds. It embeds one classical scalar Turán direction into a new mixed matrix inequality.

The other raw diagonal entry, $G_\nu = -\partial_\nu^2 \log I_\nu$, is strict log-concavity in the order and is likewise known on $(-1, \infty)$ [21, Ex. 1], [29, Thm. 3]. Order derivatives of Bessel functions have been studied independently through closed forms, integral representations, reflection identities, and large-order asymptotics [34, §10.38], [7, 15, 16, 17]. What appears not to have an antecedent in these lines is the specific Schur coupling of continuous order curvature, mixed order–argument curvature, and the corrected classical Turán direction, together with its (κ, τ) phase geometry and coefficientwise critical scaling.

8.3. The Bessel distribution and conditioning hierarchy. A recent general probabilistic approach to positive special-function series normalizes their summands into probability laws and expresses parameter log-derivatives through moments, covariances, and stochastic-order criteria, with applications including Turán inequalities [11]. That methodology is a close neighbor of the one-copy probabilistic viewpoint here; the conditioned two-copy Schur fibers, four-copy determinant sectors, and their phase geometry are additional structures specific to the present problem.

The one-copy law (4.17) is the discrete Bessel distribution studied by Yuan and Kalbfleisch and later by Devroye [44, 12]. In that language the classical scalar inequality $H_\nu > 0$ is underdispersion, $\text{Var } Y < \mathbb{E}Y$, equivalent to the lower ratio bound in [44, eq. (A.5)]. The present paper uses this law at two further levels: the coefficient fibers are two-copy Bessel laws conditioned on their total, and determinant coefficients are four-copy laws conditioned on the grand total. Power-series distributions and fixed-sum conditioning are classical [10, 35, 36]; the new content here is the resulting Bessel Schur-deficit fiber, its mixed-determinant hierarchy, and the associated phase geometry. Averaging the conditioned fibers gives the covariance-deficit matrix (4.25). We have not found this conditioned Schur-deficit hierarchy or the resulting strict digamma/count covariance inequality (4.28) stated in the discrete Bessel-distribution literature.

8.4. Total positivity and determinant hierarchies. Classical Bessel Turán determinants, Hankel determinants, and determinant monotonicity have been studied in several forms [5, 9, 20, 28]. A recent preprint of Salazar proves strict total positivity of the evaluation kernel $(z, \nu) \mapsto I_\nu(z)$ for $z > 0$, $\nu \geq 0$, extending integer-order results of Buchstaber and Glutsyuk [38, 8]; a companion preprint develops Hankel and Turán determinant hierarchies from positive moment representations [37]. These are evaluation-minor or Hankel-moment structures, not the mixed differential Schur determinant here. There is also a simple covariance obstruction to identifying our determinant with a rescaling-covariant confluent total-positivity minor. Write $F(\nu, y) = \log I_\nu(e^y)$, so $\partial_y = \Theta_z$. Replacing $I_\nu(e^y)$ by $e^{cy}I_\nu(e^y)$ leaves G_ν and P_ν fixed but sends H_ν to $H_\nu + 2c$, hence $\tilde{D}_\nu = D_\nu + 2cG_\nu$. Since $G_\nu > 0$, the sign can be changed at any fixed point by choosing c sufficiently negative. By contrast, the signs of total-positivity minors are preserved under positive row and column rescalings. The same is true for confluent y -jets: multiplication by e^{cy} sends $(K, K_y, \dots, K_{y^r})$ through an upper-triangular change of jet coordinates with positive diagonal e^{cy} , so every such confluent minor changes by a positive factor. Any canonical minor interpretation of the present inequality must therefore incorporate additional Bessel-specific normalization or nonlocal information.

8.5. Canonical–microcanonical transfer. The identity (4.33) is stronger than a numerical coincidence between coefficient and pointwise asymptotics: the Bessel determinant is the canonical average of its microcanonical determinant sectors. The concentration law (7.23) identifies the fixed total with $n \sim 2z$, while Theorems 2.9 and 2.10 show that the coefficient-wall and Bessel-wall jets agree through the first three nontrivial critical orders.

A related preprint of the author [41] proves coefficientwise Schur-concavity for cubic multiple-Pochhammer series by a beta-binomial fixed-total representation. There the degree- m coefficient of a product is a scalar microcanonical partition function for two canonical index variables conditioned on their sum, and summation over m recovers the ordinary pointwise Turán inequality. The present reciprocal-gamma problem may be viewed as a matrix-valued counterpart: the fixed-total fibers carry a two-direction Schur deficit, and determinant coefficients require the additional four-copy polarization. The two settings therefore suggest that fixed-total positivity is a reusable mechanism rather than a feature specific to the Bessel family.

This transfer is different in kind from a sharper pointwise Bessel inequality and suggests an ensemble-level mechanism that may persist for other positive special-function partition functions.

9. CONCLUDING REMARKS

The common object in this paper is a Schur deficit viewed at three resolutions. At fixed two-copy total it is a microcanonical coefficient fiber; at fixed four-copy total, mixed-determinant polarization turns those fibers into a determinant sector; and canonical mixing over the total count recovers the pointwise Bessel determinant:

$$D_{a-1}^{(\kappa, \tau)}(2\sqrt{\lambda}) = \mathbb{E}_\lambda d_N^{(\kappa, \tau)}.$$

Coefficientwise and pointwise positivity are therefore not parallel phenomena but the microcanonical and canonical positivity problems for the same discrete Bessel ensemble.

This viewpoint explains both the endpoint proof and the phase diagram. At the endpoint, the only fiber that can be indefinite is M_1 , and it can self-interact under determinant polarization only in degree two; the apparent infinite tail of possible failures collapses to one finite defect. Under the two-parameter deformation, the vacuum sector, the exposed one-particle sector, and the thermodynamic sector control the three global contacts. The remaining coefficient walls form an affine fan whose thermodynamic accumulation point sits strictly below the exposed coefficient boundary:

$$\tau_\infty(a) < \tau_{\text{cw}}(a, 1) < 1.$$

Finally, the concentration of the four-copy canonical total identifies the microcanonical degree with the Bessel argument through $n \sim 2z$. This turns the coefficient saddle into the analytic shadow of equivalence of ensembles and explains why the coefficient-wall and Bessel-wall asymptotics have the same successive critical cancellations, culminating in cubic matching at a unique point. The mechanism suggests that fixed-total positivity, finite-defect localization, and canonical–microcanonical transfer may persist beyond the Bessel family.

The following problems isolate the most direct tests of that mechanism.

Question 9.1. For the Wright-type family

$$Z_\rho(a, \lambda) = \sum_{k \geq 0} \frac{\lambda^k}{k! \Gamma(a + \rho k)}, \quad \rho > 0,$$

for which ρ do the conditioned two-copy fibers admit explicit Schur-deficit or Gram representations? When is the degree-one wall exposed over all higher sectors, and does the same canonical–microcanonical phase separation persist?

Question 9.2. Can classical mixed-discriminant positivity [3] be combined with coefficient-fiber localization so that isolated indefinite fibers affect only finitely many determinant degrees? What replaces the exceptional degree $2j$ in higher rank?

Question 9.3. For

$$\mathcal{A}_f(\theta; x) = \sum_{n \geq 0} f_n a_n(\theta) x^n, \quad f_n \geq 0, \quad a_n(\theta) > 0,$$

which kernel conditions and shape conditions on f make every nonzero fixed-total sector

$$[x^m] \mathcal{A}_f(u; x) \mathcal{A}_f(v; x)$$

Schur-concave at fixed $u + v$? When does this scalar property admit a matrix Schur-deficit lift, and when do its microcanonical wall jets match the canonical pointwise wall jets to all orders? A related preprint of the author [41] proves the scalar first step for the cubic multiple-Pochhammer kernel with nonnegative log-concave f of interval support, using beta-binomial stochastic order.

Question 9.4. Is there a canonical normalization, bordered kernel, or nonlocal minor for $(z, \nu) \mapsto I_\nu(z)$ whose positivity is equivalent to the Bessel–Schur inequality? The rescaling obstruction in Section 8.4 rules out a purely local confluent-total-positivity explanation.

Question 9.5. Can the covariance-deficit identity (4.21) be realized by a natural augmented discrete-Bessel model in which $\widehat{M}_m^{(\kappa, \tau)}$ and its canonical average are genuine observed- or Fisher-information matrices, rather than only information-deficit analogues?

APPENDIX A. THE EXPOSED ONE-PARTICLE WALL

The only lengthy algebraic verification needed for the exposed-wall argument is the boundary lemma used in Section 6. We give the details here so that the main narrative retains the geometric content of the proof.

Proof of Lemma 6.2. We establish four facts about $\hat{N}_m$.

1. *Positive definiteness for $m \geq 2$.* Apply Theorem 5.2 with $s = s_*$, so that $\hat{N}_m = N_m(s_*)$ and it suffices to show that

$$\hat{\rho}_m = \rho_m(s_*) = s_* + c_m - \left(a + \frac{m}{2} - 1\right)^2 \psi_1(a + m - 1)$$

is positive. By (5.3),

$$\hat{\rho}_m > \Lambda_m := s_* + c_m - \frac{(a + m/2 - 1)^2}{a + m - 3/2}.$$

Treating m as a real variable,

$$(A.1) \quad \frac{\partial \Lambda_m}{\partial m} = \frac{2a - 1}{2(2a + 2m - 3)^2}.$$

If $0 < a < 1/2$, then Λ_m decreases and

$$\lim_{m \rightarrow \infty} \Lambda_m = -\frac{8a^3g - 6a^2g - 8a^2 + 3}{4(2a^2g - 1)}.$$

The lower bound (5.2) makes the numerator strictly smaller than $-2a$, so the limit is positive. If $a \geq 1/2$, then Λ_m increases and its minimum is

$$\Lambda_2 = -\frac{4a^4g - 4a^3 - 2a^2g - 2a^2 + a + 1}{(2a + 1)(2a^2g - 1)}.$$

For $1/2 \leq a \leq 1/\sqrt{2}$, the coefficient of g in the numerator is nonpositive and (5.2) bounds that numerator above by $-a$. For $a \geq 1/\sqrt{2}$, the coefficient is nonnegative and (5.4) bounds it above by $-(a^2 + 1)/(3a)$. Thus $\Lambda_2 > 0$ in both cases. Hence $\hat{\rho}_m > 0$ for every $m \geq 2$, so Theorem 5.2 gives

$$(A.2) \quad \hat{N}_m \succ 0 \quad (m \geq 2).$$

2. *Pairs involving $\hat{N}_0$.* Set $F_m = \text{MD}(\hat{N}_0, \hat{N}_m)$. The definition of s_* gives $F_1 = 0$. Using $\alpha_m - \alpha_1 = -\sum_{r=1}^{m-1} (a + r)^{-2}$ and the explicit formulas for β_m, c_m , one obtains

$$(A.3) \quad F_m = gc_m - s_* \sum_{r=1}^{m-1} \frac{1}{(a + r)^2} + \frac{(a - 1)(m - 1)}{a(a + m - 1)}.$$

If $0 < a < 1/2$, then $s_* < 0$. Dropping the positive middle term and using $g > a^{-2}$ shows that $F_m > 0$ once

$$m(a + m - 1) > 2a(1 - a)(2a + 2m - 3).$$

The difference between the two sides is increasing for $m \geq 2$, and at $m = 2$ equals $2(2a^3 - a^2 + 1) > 0$.

At $a = 1/2$, one has $s_* = 0$ and, for every $m \geq 2$,

$$F_m = \frac{\pi^2 m}{8} - \frac{2(m - 1)}{2m - 1} > 0.$$

Assume now that $a > 1/2$. The telescoping estimate

$$\frac{1}{x^2} < \frac{1}{(x - 1/2)(x + 1/2)} = \frac{1}{x - 1/2} - \frac{1}{x + 1/2}$$

gives

$$\sum_{r=1}^{m-1} \frac{1}{(a + r)^2} < \frac{m - 1}{(a + 1/2)(a + m - 1/2)}.$$

Using $c_m = m(m-1)/\{2(2a+2m-3)\}$ in (A.3), it is therefore enough to prove

$$(A.4) \quad \mathcal{B}_{a,m}(g) := \frac{gm}{2(2a+2m-3)} - \frac{s_*}{(a+1/2)(a+m-1/2)} + \frac{a-1}{a(a+m-1)} > 0.$$

Put

$$\tilde{a} = a - \frac{1}{2} > 0, \quad \tilde{m} = m - 2 \geq 0, \quad g_0 = \frac{1}{a} + \frac{1}{2a^2}.$$

The lower bound (5.2) gives $g > g_0$, and therefore

$$2a^2g - 1 > 2a, \quad s_* = \frac{a(2a-1)}{2a^2g-1} < a - \frac{1}{2} = \tilde{a}.$$

Substituting these two strict bounds into (A.4) gives

$$\mathcal{B}_{a,m}(g) > \frac{\mathcal{P}(\tilde{a}, \tilde{m})}{(\tilde{a}+1)(2\tilde{a}+1)^2(\tilde{a}+\tilde{m}+1)(\tilde{a}+\tilde{m}+2)(2\tilde{a}+2\tilde{m}+3)},$$

where

$$\begin{aligned} \mathcal{P}(\tilde{a}, \tilde{m}) &= 2\tilde{a}^4\tilde{m} + 8\tilde{a}^4 + 4\tilde{a}^3\tilde{m}^2 + 23\tilde{a}^3\tilde{m} + 26\tilde{a}^3 \\ &\quad + 2\tilde{a}^2\tilde{m}^3 + 19\tilde{a}^2\tilde{m}^2 + 48\tilde{a}^2\tilde{m} + 35\tilde{a}^2 \\ &\quad + 4\tilde{a}\tilde{m}^3 + 22\tilde{a}\tilde{m}^2 + 40\tilde{a}\tilde{m} + 25\tilde{a} \\ &\quad + 2\tilde{m}^3 + 9\tilde{m}^2 + 14\tilde{m} + 8 > 0. \end{aligned}$$

Thus $F_m > 0$.

$$(A.5) \quad \text{MD}(\hat{N}_0, \hat{N}_m) > 0 \quad (m \geq 2).$$

3. *Pairs involving $\hat{N}_1$.* If $a \geq 1/2$, put $t = \psi_1(a+1)$, so $g = t + a^{-2}$. Directly,

$$(A.6) \quad \det \hat{N}_1 = \frac{(2a-1)(2a^2t-2a+1)}{4a^2(2a^2t+1)} \geq 0.$$

The second factor is strictly positive because (5.2) gives

$$2a^2t - 2a + 1 > \frac{1}{(a+1)^2}.$$

Thus $\hat{N}_1 \succeq 0$, and (A.2) plus Lemma 4.7 gives $\text{MD}(\hat{N}_1, \hat{N}_m) > 0$ for $m \geq 2$.

If $0 < a < 1/2$, then $s_* < 0$, $\beta_1 < 0 < \beta_m$ for $m \geq 2$, and $s_* + c_m > 0$. The last assertion follows from discrete monotonicity of c_m and

$$s_* + c_2 = \frac{4a^3 + 2a^2g - a - 1}{(2a+1)(2a^2g-1)} > 0$$

by (5.2). It remains to control $s_*\alpha_m - 2\beta_1\beta_m$. This quantity is positive precisely when

$$\beta_m(2a^2g - 1) > a^2\alpha_m.$$

For $a < 1$, the left side increases with m and the right side decreases, so it is enough to take $m = 2$. With $t = \psi_1(a+1)$, the difference at $m = 2$ is

$$\frac{a(2a+1) - a^2(1-a^2)t}{(a+1)^2} > \frac{a(2a^3 + 4a^2 + 2a + 1)}{(2a+1)(a+1)^2} > 0,$$

where (5.3) was applied to t . Therefore

$$(A.7) \quad \text{MD}(\hat{N}_1, \hat{N}_m) > 0 \quad (m \geq 2)$$

in all cases.

4. *The degree-two self-pair.* The only remaining possible negative contribution is the self-pair of $\hat{N}_1$. Again write $t = \psi_1(a+1)$. Direct substitution into the coefficient formula gives

$$(A.8) \quad \hat{\Delta}_2 = \frac{(a^2t+1)\mathcal{P}_2(a,t)}{\Gamma(a)^4 a^5 (a+1)^3 (2a^2t+1)},$$

where

$$\begin{aligned}\mathcal{P}_2(a, t) &= (2a^5 + 4a^4 + 2a^3)t^2 \\ &\quad + (8a^5 + 12a^4 + 5a^3 + 2a^2 + a)t \\ &\quad - 8a^4 - 6a^3 + a^2 + 4a + 2.\end{aligned}$$

This polynomial is strictly increasing in $t > 0$. At the lower bound from (5.2),

$$t > \frac{1}{a+1} + \frac{1}{2(a+1)^2},$$

one obtains

$$\mathcal{P}_2(a, t) > \frac{8a^5 + 20a^4 + 28a^3 + 30a^2 + 19a + 4}{2(a+1)^2} > 0.$$

Thus $\widehat{\Delta}_2 > 0$.

Finally, the congruence between $M_m^{(1, \tau_*)}$ and $\widehat{N}_m$ preserves the signs of all mixed determinants. The degree-one coefficient is zero because $\text{MD}(\widehat{N}_0, \widehat{N}_1) = 0$. For $n \geq 3$, every pair in the polarized coefficient formula is positive by (A.2), (A.5), and (A.7); degree two was handled separately above. This proves the lemma. $\square$

APPENDIX B. BEYOND THE POSITIVE-SERIES DOMAIN

The phase diagram and critical scaling of the preceding sections use the positive-series domain $a > 0$ in an essential way. We test what survives when the order crosses its endpoint $\nu = -1$. The behavior changes qualitatively almost immediately. On the open set

$$\mathcal{U} = \{(\nu, z) \in (\mathbb{R} \setminus \{-1, -2, \dots\}) \times (0, \infty) : I_\nu(z) \neq 0\},$$

define G_ν , P_ν , H_ν , and D_ν by (2.11)–(2.15) with $\log I_\nu$ replaced by $\log |I_\nu(z)|$; the order derivatives are taken on $\mathcal{U}$, and for each fixed ν the resulting formulas restrict to any connected component of $\{z > 0 : I_\nu(z) \neq 0\}$. Where $I_\nu > 0$ this agrees with the ordinary logarithm.

For every real $a \notin \{0, -1, -2, \dots\}$, the partial-fraction expansion

$$(B.1) \quad \psi_1(a) = \sum_{r=0}^{\infty} \frac{1}{(a+r)^2}$$

holds and shows that $g = \psi_1(a) > 0$ [33, eq. 5.15.1].

Lemma B.1. *For each fixed coefficient index, the scalar coefficient identities in (4.2)–(4.7), and every resulting finite-convolution formula for a coefficient of Δ , extend meromorphically in a . A formula containing the product $S_m c_m^{(\kappa)}$ is understood as that complete product; its apparent pole at $2a + 2m - 3 = 0$ is removable. If $a \in \mathbb{R} \setminus \{0, -1, -2, \dots\}$, $\lambda > 0$, and $Z(a, \lambda) \neq 0$, then (2.20), (2.18), and (2.21) remain valid, with the positive real square root of $\psi_1(a)$ and with logarithmic derivatives interpreted through $\log |Z|$ and $\log |I_{a-1}|$ on the relevant nonvanishing component.*

Proof. By Lemma 2.1, Z and all fixed-order derivatives used here are entire in (a, λ) . For each fixed coefficient index, both sides of each scalar coefficient identity are meromorphic functions of a and agree for $a > 0$; the one-variable identity theorem extends them throughout their common domain. Each coefficient of $\Delta = AC - gB^2$ is a finite convolution of such scalar coefficient functions and is therefore meromorphic as well. The zero of S_m cancels the apparent pole in $S_m c_m^{(\kappa)}$.

On the real nonpole set, (B.1) gives $g = \psi_1(a) > 0$, so the positive real square root used in the matrix formulas is well defined. On each connected component where $Z \neq 0$, Z has constant sign and $\partial_a \log |Z| = Z_a/Z$, so the differential identities remain valid. Finally, (2.10) gives

$$\log |I_{a-1}(2\sqrt{\lambda})| = \frac{a-1}{2} \log \lambda + \log |Z(a, \lambda)|,$$

which transfers the same identities to the Bessel side. $\square$

Away from the poles the leading small- z term keeps its positive sign. For $a = \nu + 1 \in \mathbb{R} \setminus \{0, -1, -2, \dots\}$ the ascending series gives $I_\nu(z) = (z/2)^\nu \Gamma(\nu + 1)^{-1} (1 + O(z^2))$, so I_ν has constant sign near $z = 0$; by Lemma B.1, (5.13) and (2.21) with $Z(a, 0) = 1/\Gamma(a)$ and $\lambda = z^2/4$ give

$$(B.2) \quad D_\nu(z) = \frac{2(a\psi_1(a) - 1)}{a^3\psi_1(a)} z^2 + O(z^4) \quad (z \downarrow 0),$$

whose coefficient is positive for every such a : for $a > 0$ by (5.2), and for negative nonintegral a because (B.1) makes $a\psi_1(a) - 1$ and $a^3\psi_1(a)$ both negative. Coefficientwise positivity of Δ nonetheless fails already on the next interval where $1/\Gamma(a) > 0$: for $-3 < \nu < -2$ one has $a \in (-2, -1)$ and $\Gamma(a) > 0$.

Proposition B.2. *If $-2 < a < -1$, equivalently $-3 < \nu < -2$, then*

$$(B.3) \quad [\lambda^2]\Delta(a, \lambda) = \Delta_2(a) < 0.$$

Thus the coefficientwise positivity theorem for Δ cannot be extended to that interval.

Proof. By Lemma B.1, the identity (5.15)–(5.16) persists on $-2 < a < -1$. Set $b = -a \in (1, 2)$, so that $a + 1 \in (-1, 0)$. The denominator in (5.15) is then negative: $a^6 > 0$ and $\Gamma(a)^4 > 0$, while $(a + 1)^3 < 0$. It therefore suffices to show $Q_*(a, t) > 0$. By (B.1), $t = \psi_1(a + 1) > 0$ and

$$\mathcal{R}_a = t - \frac{1}{a + 1} > \frac{1}{b - 1}.$$

The first term in (5.16) is positive. The second and third terms satisfy

$$\begin{aligned} & 2b^2(b - 1)^2(8b^2 - 3b + 1)\mathcal{R}_a - 2b(b - 1)(5b - 3) \\ & > 2b(b - 1) [8b^3 - 3b^2 - 4b + 3] > 0. \end{aligned}$$

Indeed, the polynomial in brackets equals 4 at $b = 1$ and has derivative $24b^2 - 6b - 4 > 0$ for $b \geq 1$. Hence $Q_*(a, t) > 0$, proving (B.3). $\square$

A complete classification beyond $\nu > -1$ appears to require separate treatment.

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